A Hybrid DistilBERT–Logistic Regression Model for Taglish Misleading Information Detection Using Contextual and Linguistic Features

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CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Introduction

The increasing prevalence of misleading information has become a significant challenge in the Philippine digital information environment, where political discussion, public debate, and everyday information sharing increasingly occur through social media platforms, comment sections, group chats, short-form videos, and other user-generated online spaces.

In contemporary information environments, misleading content can spread quickly and influence how people understand public issues, evaluate institutions, and participate in civic discourse. In the Philippine context, this concern is particularly urgent because, as reported by the Philippine Daily Inquirer and published by Tsek.ph (2022), a survey released by the Social Weather Stations revealed that seven out of ten adult Filipinos believe that fake news in media is a serious problem, while many respondents also consider fake news on social media platforms such as Facebook, Twitter, and YouTube to be a major concern.

Accordingly, misleading-information detection has become an important area of computational research. Automated systems can assist in examining large amounts of online content and identifying textual patterns associated with false, inaccurate, fabricated, or materially misleading claims. However, such systems cannot reliably determine whether an author intentionally intended to deceive based on text alone. Therefore, the present study treats misleading-information detection as a supervised text-

classification task based on verified and annotated examples rather than as a method for judging the intention of a content creator.

Beyond its technical relevance, misleading information also presents broader social and developmental concerns. Access to reliable information supports informed decision-making, public participation, and trust in institutions. The proposed study is related to Sustainable Development Goal 16: Peace, Justice and Strong Institutions, particularly Target 16.10, which emphasizes public access to information and the protection of fundamental freedoms. It is also related to Sustainable Development Goal 4: Quality Education because encouraging users to examine questionable online information may support digital literacy, critical thinking, and responsible information use (United Nations, 2015). Nevertheless, the proposed system does not claim to achieve these goals independently.

Instead, it may contribute by providing users with an additional tool for recognizing textual content that requires further verification.

Developing such a system for the Philippine setting requires consideration of the language commonly used by Filipino internet users. Online communication frequently contains Taglish, in which Tagalog and English are combined within the same statement or conversation. Taglish content may include code-switching, Filipino particles, informal spelling, abbreviations, slang, and culturally specific expressions. Herrera et al. (2022) described Tagalog-English code-switching as a common form of discourse in Philippine culture while identifying Tagalog and Taglish as underrepresented areas in natural language processing research.

Consequently, models developed primarily from formal or monolingual text may have difficulty representing mixed-language content whose meaning depends on local expressions, switching patterns, and surrounding context.

Existing studies provide an important foundation but do not completely address the problem considered in the present research. Cruz et al. (2020) developed Fake News Filipino and demonstrated that transfer-learning approaches can support fake-news classification in a low-resource Philippine language setting. However, the dataset primarily contains Filipino news articles and was not specifically designed for informal Taglish posts. Conversely, the TweetTaglish dataset developed by Herrera et al. (2022) represents Tagalog-English code-switching but was designed to investigate code-switching rather than classify misleading information. Thus, a research gap remains in developing and evaluating a model specifically intended for potentially misleading Taglish textual content encountered through social-media posts and online articles.

To address this gap, the present study proposes A Hybrid DistilBERT–Logistic Regression Model for Taglish Misleading Information Detection Using Contextual and Linguistic Features. DistilBERT will generate contextual representations of the submitted text, while seven explicit feature groups will measure code-switching patterns, Filipino particles, punctuation intensity, capitalization patterns, repeated characters, emotional-language cues, and urgency cues. These representations will be combined and supplied to Logistic Regression for binary classification. The explicit features will serve only as measurable textual indicators and will not be considered individual proof that the content is not classified from any single feature alone. The four model configurations will be compared using Macro F1 as the primary measure, supported by accuracy, class-level

precision, recall and F1-score, confusion matrices, false-positive rate, and inference time. The selected model will be integrated into a Google Chrome extension that analyzes user-selected text or captions from supported social-media posts and text entered manually. The system may retrieve related credible sources as references for review, but retrieval does not alter the classifier prediction.

Background of the Study

The rapid growth of digital platforms has made information easier to access and share. However, the same accessibility has also made false and misleading information more difficult to identify and control. Social media platforms, news websites, and video-sharing platforms allow information to reach large numbers of users within a short period, sometimes before its accuracy can be verified. This situation is particularly relevant in the Philippines, where social media continues to play an important role in how people access news and public information.

According to the Reuters Institute's Digital News Report 2026, social media remains a dominant source of news in the country, while overall trust in news has declined to 28%. The report also noted that users increasingly encounter news through platform feeds affected by political messaging, polarization, and disinformation (Reuters Institute for the Study of Journalism [RISJ], 2026). These conditions highlight the need for tools that can assist users in identifying information that may require further verification.

Misleading information is not limited to fabricated news articles. It can also appear in social-media posts, online articles, screenshots, images containing text, and videos containing spoken or displayed claims. UNESCO (2024) distinguishes misinformation from disinformation mainly through intent. Misinformation refers to false information

shared without an intention to deceive, while disinformation involves an intention to deceive or manipulate. However, determining the intention of the original creator based only on the content is difficult. Because of this limitation, the present study focuses on classifying content as Misleading or Non- Misleading rather than determining whether the creator intentionally produced disinformation. This approach allows the proposed system to focus on the characteristics and available evidence of the information itself.

The Philippine information environment also presents challenges related to the way misleading information is produced and shared. Previous studies have documented the organized use of online platforms for political communication and the spread of coordinated narratives, particularly during election periods. Arugay and Mendoza (2022) discussed the use of coordinated online activities, financial resources, and influencer networks in shaping political narratives and public perception during the 2022 national elections. These findings suggest that misleading information may be presented in ways that are intended to appear credible, familiar, or persuasive to Filipino audiences. Therefore, an automated detection system should not depend only on specific words or keywords but should also consider the context and linguistic characteristics of online content.

Language is an additional challenge in detecting misleading information in the Philippine setting. Filipino users commonly communicate through a combination of Filipino and English, often referred to as Taglish. Tagalog-English code-switching is widely observed in Philippine online communication, but it remains relatively underrepresented in natural language processing (NLP) research. Herrera et al.

(2022) introduced TweetTaglish, a dataset containing more than 20,000 instances for studying Tagalog-English code-switching and demonstrating the need for NLP resources that reflect actual Filipino online communication. More recent Filipino NLP research has also emphasized the limited representation of Filipino and code- switched Taglish in existing language resources (Montalan et al., 2025). This presents a challenge for automated systems because online Filipino content may contain mixed-language expressions, informal spelling, slang, and other language patterns that may not be sufficiently represented in models developed primarily from high-resource languages.

Previous research has shown that machine learning and NLP can be applied to the detection of false or misleading information in Philippine content. Cruz et al.

(2020) developed Fake News Filipino, an expertly curated dataset containing 3,206 Filipino news articles divided equally between real and fake articles. Their study demonstrated the potential of transfer-learning approaches for fake-news classification in a low-resource language, achieving 91% accuracy in their experiments. Other resources, such as the Philippine Fake News Corpus, have also been developed to examine linguistic characteristics that may be useful in fake-news detection. These resources provide a useful starting point for the present study. However, differences in their sources, labeling methods, language characteristics, and content types mean that existing data cannot simply be assumed to be suitable as final training data.

For this reason, the quality and reliability of the dataset are important parts of the proposed study. The system will use an evidence-based dataset development process in which candidate samples are screened for language and quality, checkable claims are identified, supporting or contradicting evidence is retrieved, and samples are reviewed by

human validators. Samples with insufficient evidence, missing context, unresolved disagreement, or non-checkable claims will be excluded rather than being forced into either classification. Existing Philippine datasets, including Fake News Filipino, BalitaNLP, NewsPH, the Philippine Fake News Corpus, and multilingual Facebook comments, will therefore serve as potential sources of candidate samples that must undergo appropriate validation before being included in the final dataset. This process is intended to reduce unreliable labels and provide a stronger foundation for model training and evaluation.

Another concern is that a machine-learning model may learn patterns that are specific to the dataset instead of characteristics that are actually related to misleading information. For example, a model may associate certain publishers, sources, or writing styles with a particular classification rather than examining the content itself. Przybyla (2020) demonstrated that text classifiers can overfit to sources represented in their training data, highlighting the importance of source- independent testing. Linguistic and stylistic characteristics, including readability, syntactic patterns, sentiment, and writing style, may also provide useful information for classification (Choudhary & Arora, 2021; An empiric validation of linguistic features in machine learning models for fake news detection, 2023). To address these concerns, the proposed study will incorporate measurable linguistic and stylistic features while controlling duplicates, grouping related claims, auditing sources, and preventing publisher or source identity from being directly used as classifier features.

To address the linguistic and contextual challenges, the study proposes a hybrid machine-learning model that combines multilingual DistilBERT, linguistic and stylistic

features, and Logistic Regression. Multilingual BERT-based models can generate contextual representations across multiple languages and support cross-lingual transfer, although their effectiveness may vary depending on the language and available training resources (Pires et al., 2019; Wu & Dredze, 2020).

DistilBERT is a smaller and faster version of BERT that retains much of its language-understanding capability while requiring fewer computational resources (Sanh et al., 2019). In the proposed model, multilingual DistilBERT will provide contextual representations of the input, while linguistic and stylistic features will provide additional measurable characteristics of the content. These features will then be combined and processed using Logistic Regression to classify the content as Misleading or Non-Misleading.

The proposed system also considers the different forms through which users encounter information online. Instead of requiring users to manually copy and paste content into a separate application, the study proposes a browser extension that can inspect supported webpage elements through the Document Object Model (DOM) and Shadow DOM. The system is designed to process readable webpage text, images or screenshots containing readable text, and supported video content. Optical Character Recognition (OCR) will be used to extract text from images and screenshots, while supported videos will undergo speech-to-text processing to obtain transcripts that can be passed to the classification pipeline.

This multimodal approach reflects the increasing use of visual and video-based content as sources of online information. The Reuters Institute has also reported the

continued growth of social video consumption, including in the Philippines (Newman et al., 2025).

The proposed system is intended as a screening and decision-support tool, not as a replacement for professional fact-checkers or an absolute authority on factual truth. Its purpose is to help users identify content that may require further verification. Instead of presenting its results as definitive judgments, the system will use cautious classifications such as Potentially Misleading and No Misleading Information Detected. The results may also include a confidence score, contextual information, credible-source references where available, and an appropriate disclaimer. Since automated classification may produce both false positives and false negatives, the study will evaluate the system using accuracy, precision, recall, F1-score, confusion matrix, and False Positive Rate. The proposed hybrid model will also be compared with baseline approaches, including TF-IDF with Logistic Regression, multilingual DistilBERT with Logistic Regression, and linguistic and stylistic features with Logistic Regression.

Ultimately, the study aims to develop a practical and evidence-based approach to identifying potentially misleading information in the Philippine online environment. The proposed system combines validated dataset development, human verification, multilingual NLP, linguistic and stylistic analysis, multimodal text extraction, and browser-based screening. A minimum target of 4,000 validated Taglish samples, supplemented by validated Filipino and English data, will be developed for model training and evaluation. The study will further incorporate controlled dataset and model maintenance, model versioning, and authorized retraining rather than allowing the system to automatically learn from unverified public submissions. Through these components, the study aims to

provide Filipino internet users with a practical tool for screening online information while maintaining the important principle that automated results should be treated as an aid to verification rather than a final determination of truth.

Project Context

SuriPH will be developed as a browser-based screening prototype for selected supported social-media post content, with Taglish as its primary language domain and Filipino and English as supplementary groups. SuriPH is a proper system name, not an acronym.

The public-facing component will operate through a React-based Google Chrome extension. Browser-side content scripts will inspect only user-selected text or captions from supported social-media post elements that are accessible under granted permissions. The prototype will not claim compatibility with other browsers or unrestricted extraction of complete articles, entire feeds, closed shadow roots, or protected page structures. Manual text entry will remain available when direct extraction is unsupported.

The extension will send only the resulting textual content or approved media-processing request to the backend. The Python and FastAPI backend will coordinate text preparation, EasyOCR processing, Whisper transcription, contextual representation, linguistic and stylistic feature extraction, Logistic Regression inference, independent credible-source retrieval, database operations, file handling, and model-version management. This separation keeps model artifacts, database credentials, and third-party service keys outside browser-side code.

Content scripts will inspect only supported page elements that the browser extension can access with the required permissions. This may include regular DOM

elements and accessible or open Shadow DOM. Closed shadow roots, protected media, cross-origin restrictions, and platform changes may prevent acquisition. These conditions will produce an appropriate status or failure record instead of an unsupported compatibility claim.

Image and screenshot support will rely on EasyOCR to convert visible words into machine-readable text. The OCR result may contain recognition errors caused by small fonts, decorative type, low contrast, compression, rotation, mixed scripts, or cluttered backgrounds. The prototype will therefore treat OCR as a text- acquisition stage and will measure extraction quality independently from classifier quality. A wrong model prediction and an incorrect OCR transcript represent different failure sources and must be evaluated separately.

For supported webpage videos, Whisper will obtain spoken transcripts from accessible audio, while EasyOCR will process selected frames for visible on- screen text. The transcript and relevant frame text will be combined after normalization and deduplication. The system will not examine facial identity, object authenticity, editing patterns, scene semantics, or manipulated media. Its video capability is limited to obtaining textual information for the same classifier used by the other supported inputs.

Purpose and Description

This study sets out to build and test a hybrid DistilBERT-Logistic Regression model that can flag misleading information in Taglish online content. A central question drives the design: does fusing contextual embeddings with explicit linguistic and stylistic features actually classify better than either approach used alone, or is that just an assumption worth testing? Philippine research on this problem has grown along two tracks that never

quite met. Cruz et al. (2020) put together the first well-curated Filipino fake-news corpus and proved transfer learning holds up even with scarce labeled data, but their material is formal news writing with a plain real-or-fake label attached. Herrera et al. (2022) took the opposite angle, releasing a large corpus of Tagalog-English code-switched posts, though theirs was built to study code-switching itself and carries no veracity labels whatsoever. Neither project reaches into spoken content, even as more and more political and public-health claims in the Philippines now travel as short-form video rather than text. Matos et al. (2026) offer a way through this: false claims inside videos can be located simply by classifying the audio transcript, no visual analysis needed. Taking that cue, this study folds written Taglish posts and machine- generated Taglish video transcripts into a single classification pipeline.

At the model level, the study uses the binary research labels Misleading and Non-Misleading. Misleading refers to a checkable claim that conflicts materially with the accepted evidence under the annotation codebook; Non-Misleading refers to a checkable claim supported by the accepted evidence. Non-checkable, context-deficient, unverifiable, or unresolved records are excluded instead of being forced into either class. In the public interface, these classes are presented cautiously as Potentially Misleading and No Misleading Information Detected so that model predictions are not presented as definitive factual judgments. The system does not infer the author’s intent.

Non-Misleading content is whatever reads as factual, neutral, informational, or simply opinion, with no deceptive claims baked in. Misleading content, on the other hand, is whatever turns out false, fabricated, manipulated, or unsupported once checked against credible fact-checking references or verifiable evidence. Because the label rides

on what the evidence says rather than on how an annotator personally feels about the wording, agreement scores end up measuring something specific: how consistently the team applies the evidence-based codebook, not how similarly they each judge truth.

On the dataset side, the target is a minimum of 4,000 approved Taglish samples, split evenly at 2,000 Misleading and 2,000 Non-Misleading, spread across eight topic categories at roughly 500 samples each and 250 per label within a topic: Politics/Governance/Elections, Health/Medicine, Economy/Finance/Employment, Public Safety/Disaster/Security, Science/Technology/Environment, Education/Social Issues, Entertainment/Public Figures, and Other/General. Since not every candidate will survive annotation and quality screening, the plan starts with a wider net, 5,500 to 6,000 raw candidates, and collection keeps going until 4,000 samples have genuinely cleared screening rather than stopping the moment the raw count hits its target.

Labeling itself follows an evidence-grounded pipeline instead of leaning on researcher judgment. Three researchers agreeing a claim is false doesn't actually make it false, so the process works differently: each candidate's claim gets extracted, then checked against outside evidence. A retrieval module built on SearXNG or Serper pulls credible source links when a post looks legitimate or just needs a general check, while the Google Fact Check Tools API steps in for anything misleading or flagged for review, returning the full fact-check card, claim text, claimant, publisher, review URL, and rating. Annotators then match the claim to what that evidence shows rather than going with a gut read. Samples with no locatable evidence get excluded outright instead of being forced into a label. Fleiss' Kappa still measures inter-rater reliability, only now it's measuring how

consistently the evidence-based protocol gets applied; a kappa of 0.61 or higher is the floor, matching the substantial-agreement threshold from Landis and Koch (1977).

One more safeguard sits in the annotation step: every sample gets a claim group identifier, so that near-duplicate versions of the same underlying claim stay grouped together and land in a single split rather than leaking across training, validation, and testing. Once the 4,000-sample dataset is approved, it splits 70/15/15 into 2,800 training, 600 validation, and 600 testing samples, keeping the 50/50 label balance and topic spread as close as possible in each. Training is what the model actually learns from. Validation supports decisions made along the way, picking a classification threshold, comparing configurations, without ever touching the test set. Testing stays untouched until the very end, run exactly once after the model and threshold are locked, and those are the numbers that get reported.

On the deployment side, the whole thing lives as a browser extension backed by a machine learning service, built first for Google Chrome under Manifest V3.

Picture the ordinary case: someone reading a Facebook post or a comment thread highlights a passage and asks for a check. For video, the flow shifts slightly, the user triggers the extension mid-clip, the extension grabs a bounded chunk of the tab's audio, transcribes it, and sends that Taglish transcript down the same path as written text. Either way, what comes back is a predicted label, a confidence score, and the specific linguistic indicators that pushed the result, shown to the user as something like a warning that the content looks potentially misleading, alongside flagged indicators such as urgency language, heavy punctuation, or emotional wording. That warning is framed plainly as an automated screening result, not a verdict, with a nudge to verify important claims through

credible sources. Nowhere does the system claim someone intentionally created misleading content; it only reports that the content matches known patterns.

Alongside the Chrome extension, the prototype also includes a separate admin website, built and hosted apart from the extension, giving the research team a place to manage the dataset, review annotation and evidence records, watch dataset balance across topics and labels, and sign off on retraining. AFP administrators, as the study's beneficiary, are given a limited, view-only account on the same website scoped to the performance and quality information relevant to their ISO/IEC 25010 evaluation, rather than dataset or retraining controls.

Retraining never fires automatically off raw user submissions. It runs on explicit drift, performance, and volume triggers, and any candidate model that comes out of retraining gets compared against whatever is currently deployed and needs researcher approval before it goes live. That keeps the whole system semi- automated rather than either fully manual or, worse, fully automatic: machines handle preprocessing, duplicate detection, Taglish screening, claim extraction, topic suggestion, and evidence retrieval, while people verify evidence matches, settle disagreements, audit dataset balance, and approve every model version before deployment.

The Information Technology Department of the Armed Forces of the Philippines at Camp Aguinaldo, Quezon City, stands as the prototype's designated beneficiary and technical evaluator. Their interest is operational, they deal daily with open- source information and want assistive screening tools for it, and their staff bring the technical background needed to judge whether the system holds up against professional software standards, assessed through an ISO/IEC 25010:2023- based evaluation.

Performance gets judged against something, not in a vacuum. The hybrid model will be measured against two reference configurations, a TF-IDF baseline paired with Logistic Regression, and a DistilBERT-only model that skips the linguistic and stylistic features entirely. That comparison is what makes it possible to actually say what the fusion contributes, rather than just assuming it helps.

Reporting covers accuracy, precision, recall, F1-score, a confusion matrix, and inference time, plus a topic-by-topic breakdown, since a single overall number can hide a topic where performance is genuinely weak. Text-based and transcript- based results get reported separately too, since transcription error is a known way accuracy degrades in lower-resourced languages.

Two boundaries shape the whole design. First, punctuation, capitalization, and urgency phrasing are just ordinary features of informal Filipino writing, not proof of falsehood on their own, so Misleading and Non-Misleading samples get pulled from comparable registers and platforms. Sourcing Misleading content from viral posts while pulling Non- Misleading content from formal news articles would let the model cheat by learning writing style instead of veracity, so the study deliberately avoids that setup. Second, a screening tool that over-flags legitimate content costs something real to public discourse, not just to an accuracy score, so false positives against Non-Misleading content get their own dedicated reporting rather than disappearing into a single error figure. Every output is framed as a prompt for review, never a verdict, and the analytics view keeps the linguistic basis of each prediction inspectable. The system only acts when asked: audio gets captured solely on explicit user action, only for a bounded clip segment, and gets discarded the moment transcription finishes. Nothing gets removed, no accounts

get reported, no browsing gets monitored, and the system never claims authority over what's actually true. A predicted label is just a prompt to look closer, final judgment stays with the reader and with qualified human fact-checkers.

Conceptual Framework

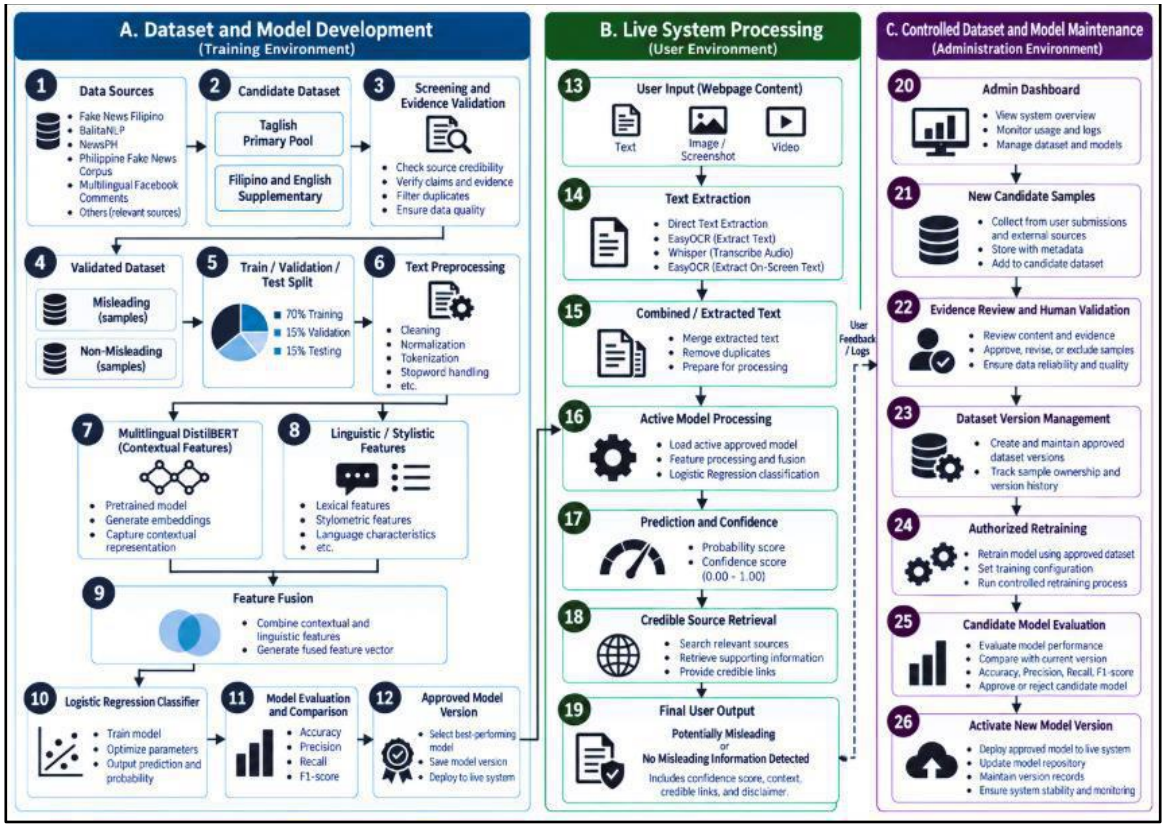


Figure 1.1 Conceptual Framework

The conceptual framework contains three connected areas: Dataset and Model Development, Live System Processing, and Controlled Dataset and Model Maintenance. The separation clarifies which activities occur before deployment, which occur during a user's screening request, and which require administrative authorization.

The conceptual framework begins with Dataset and Model Development. Candidate sources are inventoried with their provenance and original purpose. Records then pass through language screening, claim identification, evidence retrieval, human

validation, duplicate review, claim grouping, and dataset-version approval. Only approved records move to partitioning and model development. This sequence treats dataset construction as a documented research activity rather than a preliminary technical task.

Every model configuration will use the same approved claim-group-aware partitions. Vocabulary construction, scaling, and other fitted transformations will use training data only. Training and validation data will support model selection; the held-out test set will be reserved for the final evaluation. Where Figure 1 abbreviates this sequence, partitioning takes precedence over fitting any data- dependent transformation.

An explicit user request starts live processing of selected supported social- media post content. The extension will inspect permitted page structures and send the accessible text or approved media-processing request to the backend. OCR and transcription will supply textual input where required. After text preparation, contextual and linguistic features will be fused and Logistic Regression will return the internal class and model probability.

The presentation stage combines information without merging its evidentiary meaning. The prediction, confidence value, disclaimer, system status, and related sources appear in one interface, but the classifier and retrieval service remain independent. This separation allows the user to compare the automated screening result with external context and avoids the misleading impression that a retrieved link mathematically determined the prediction.

The restricted Admin Dashboard will support candidate and evidence review, Approve, Revise, or Exclude decisions, approved dataset versioning, authorized

retraining, and candidate-model evaluation. Manual approval or rejection is separate from activation: only an approved model version may become active.

Public submissions remain candidates, and the model does not change merely because new requests or submissions arrive.

Objectives

The general objective of this study is to develop and evaluate a hybrid machine-learning model that combines DistilBERT contextual embeddings with Logistic Regression and selected linguistic and stylistic features for detecting misleading information in Taglish text and to implement it as a functional Google Chrome extension.

Specifically, this study aims to:

1. Prepare an evidence-annotated dataset of at least 4,000 Taglish samples, targeting balanced Misleading and Non-Misleading classes across eight topics and using claim-group-aware partitions.

2. Develop and compare four Logistic Regression configurations using TF-IDF, multilingual DistilBERT contextual representations, seven explicit linguistic and stylistic features, and their proposed fusion.

3. Evaluate the configurations using Macro F1 as the primary metric, supported by accuracy, class-level precision, recall and F1-score, confusion matrices, false-positive rate, inference time, and paired McNemar comparisons on the held-out test set.

4. Develop a Google Chrome extension that analyzes user-selected supported post text or captions and manually entered text, displays cautious classification labels and model confidence, and retrieves related credible sources independently of classification.

5. Evaluate the implemented prototype through functional and integration testing and separate technical and general-user assessments based on selected applicable ISO/IEC 25010:2023 characteristics.

Significance of the Study

This study offers direct contributions to the following groups.

Filipino Social Media Users. Ordinary Filipino users who browse social media on Google Chrome gain access to an assistive tool that can flag potentially misleading Taglish content, whether encountered as text or as short-form video, in real time. The system provides a classification label and the specific linguistic features that triggered the result, helping users make more informed decisions before sharing or acting on content without requiring any technical expertise.

Fact-checking and Media Literacy Groups. Organizations such as Vera Files and Tsek.ph face a volume problem in reviewing misleading content. This system can serve as a preliminary screening tool that filters high-risk Taglish text and video content for closer human review, reducing manual workload without replacing the judgment of professional fact-checkers.

Software Developers and NLP Practitioners. The full pipeline, from Whisper- based transcription and calamanCy preprocessing to DistilBERT feature extraction, feature fusion, Logistic Regression classification, FastAPI deployment, and Chrome extension implementation, is documented in enough detail to support reproduction and adaptation for related problems.

Future Researchers. The methodology, evidence-grounded dataset, and pipeline documented in this study provide a reproducible reference for future research on Taglish NLP, Filipino misleading-information detection, and hybrid machine-learning systems for low-resource languages.

Students and Academic Institutions. This study offers a feasible, measurable, and locally relevant machine learning capstone project that combines social relevance, technical implementation, and standard software quality evaluation.

AFP IT Department, Camp Aguinaldo, Quezon City. The AFP IT Department serves as the primary technical evaluator and institutional beneficiary of this study.

Selected IT personnel from this department will assess the Chrome extension prototype using an ISO/IEC 25010:2023-based questionnaire covering functional suitability, performance efficiency, interaction capability, reliability, security, and maintainability. Their evaluation provides technically grounded feedback that validates the system's quality as a research-grade prototype.

Scope and Limitations of the Study

This study covers the design, development, training, testing, deployment, and evaluation of a Chrome extension-based Taglish misleading-information detection prototype. The language scope is limited to Tagalog-English code-switched text.

Purely English, purely Filipino, and other Philippine-language combinations are outside the primary scope. A candidate sample must contain sufficient Tagalog and English linguistic content to meet the study’s operational Taglish criteria.

The dataset will contain a minimum of 4,000 approved Taglish samples, consisting of 2,000 misleading and 2,000 non-misleading samples. Approximately 500 samples will be collected for each of the eight topic categories, with approximately 250 samples per class within each category. The dataset will be divided into 2,800 training, 600 validation, and 600 testing samples. Samples expressing the same claim or close paraphrases of the same claim will remain in only one split to prevent information leakage. The study will use public and ethically usable content. It excludes private messages, restricted accounts, passwords, login information, private browsing data, and unnecessary personally identifying information.

The annotation process will involve four researchers. All four will independently annotate a pilot set of approximately 100 to 150 samples. Full annotation will use a primary annotator, secondary verifier, and adjudication procedure. Unresolved or unverifiable samples will be excluded. The prediction task is limited to two classes: misleading information and non-misleading information. The system will not determine whether an author intentionally attempted to deceive another person.

The system is primarily limited to text. Video content may be processed only through its available or generated text transcript. The system will not examine images, facial expressions, gestures, background music, visual editing, voice characteristics, or deepfake indicators. The system will not conduct live fact- checking against the entire internet whenever a prediction is requested. Its output will represent a classification based on patterns learned from the approved training dataset. Therefore, its prediction cannot be treated as independent confirmation that a new claim is objectively true or false.

The displayed confidence is the model's estimated probability for its predicted class under the fitted classifier. It is not an accuracy percentage, a probability that a statement is objectively true or false, or a substitute for verification. The interface will therefore use the label Model confidence and will pair the result with a limitation notice and related sources for independent review.

Deployment will be limited to Google Chrome using Manifest V3 and the minimum permissions needed for user-activated text analysis. The extension will not automatically monitor all browsing activity. It will not remove content, block websites, report accounts, collect credentials, or identify the person responsible for a post.

The administrative module will be limited to authorized dataset management, annotation review, performance evaluation, prediction-log review, model versioning, and controlled retraining. User-submitted content will not automatically modify the trained model.

Model performance will remain dependent on the size, quality, topic distribution, language distribution, and time coverage of the dataset. Sarcasm, satire, irony, newly developing events, unfamiliar claims, spelling variation, emerging slang, and missing context may reduce classification performance.

Finally, the system is an assistive screening tool. It is not intended to replace professional fact-checkers, journalists, subject-matter experts, or the critical judgment of users.

Definition of Terms

Annotation. Refers to the process of assigning appropriate labels to eligible text samples based on documented criteria, supporting evidence, and the annotation procedure.

Binary classification. Refers to the process of assigning text to one of two categories: misleading information or non-misleading information.

Code-switching. Refers to the use or alternation of two or more languages within the same communication, sentence, phrase, or word.

Confidence Score. Refers to the numerical value produced by the classifier to indicate its estimated confidence in the predicted category. It does not represent the exact probability that the information is objectively true or false.

Contextual embeddings. Refer to numerical representations of language that capture the meaning of words based on the surrounding words and sentence context.

DistilBERT. Refers to the lightweight transformer-based model used to generate contextual representations of the processed text before classification.

DOM inspection. Refers to the process of accessing relevant textual content contained within webpage elements for subsequent screening by the system.

Feature extraction. Refers to the process of obtaining relevant and measurable characteristics from raw data for computational analysis or machine-learning applications.

Feature fusion. Refers to the combination of different types of information or representations into a unified representation for analysis or classification.

Filipino Particles. Refer to Filipino linguistic elements occurring in Taglish text that are identified and measured as part of the explicit linguistic feature set.

Hybrid Model. Refers to the model configuration that combines DistilBERT contextual representations with selected linguistic and stylistic features before classification through Logistic Regression.

Linguistic features. Refer to measurable characteristics of Taglish text, particularly code-switching patterns and Filipino particles, that are extracted as part of the explicit feature representation.

Logistic Regression. Refers to the classification algorithm used to process the combined representation and determine whether the input belongs to the misleading or non-misleading category.

Machine learning. Refers to a computational approach in which systems learn patterns from data and use the learned patterns to perform tasks such as classification or prediction.

Misleading information. Refers to information containing false, inaccurate, incomplete, or deceptive claims that may cause users to develop an incorrect understanding of an issue.

Natural Language Processing. Refers to the computational processing and analysis of human language using techniques that enable computers to represent and interpret linguistic information.

OCR, or Optical Character Recognition. Refers to the process of extracting textual information from supported images, screenshots, and selected video frames for subsequent text processing.

Punctuation intensity. Refers to the measurable use and frequency of punctuation marks within the input text as one of the explicit stylistic features.

Repeated characters. Refer to consecutive repetitions of letters or other characters within Taglish text, measured as one of the explicit stylistic features.

Shadow DOM inspection. Refers to the process of accessing supported textual content contained within encapsulated webpage structures for subsequent screening.

Speech transcription. Refers to the conversion of spoken content from supported videos into written text before the resulting text is processed by the detection pipeline.

Stylistic Features. Refer to observable characteristics of written language represented as measurable values. These include punctuation intensity, capitalization patterns, repeated characters, emotional-language cues, and urgency cues.

Taglish. Refers to communication that combines Tagalog and English linguistic elements within the same text or interaction. It may include Filipino particles, English vocabulary, local expressions, abbreviations, and other characteristics of Filipino-English communication.

Taglish filtering. Refers to the process of identifying and retaining candidate samples containing both Filipino and English linguistic components for dataset preparation.

Transformer Model. Refers to a neural-network architecture designed to process language by capturing relationships between words and their surrounding context.

Urgency Cues. Refer to expressions or wording that communicate urgency or immediacy and are represented as one of the explicit stylistic features.

CHAPTER 2

REVIEW OF RELATED LITERATURES AND STUDIES

This chapter reviews foreign and local literature and studies relevant to Taglish misleading-information detection. It establishes the linguistic, computational, dataset, and evaluation foundations of the study and identifies the research gap addressed by the proposed system.

Review of Related Literature

Foreign Literature

The widespread use of online platforms has increased the speed and scale at which inaccurate and misleading information can circulate. Aïmeur et al. (2023) explained that misinformation, disinformation, and fake news are related forms of problematic information but differ in characteristics such as accuracy and intention.

Broda and Strömbäck (2024) similarly observed that definitions of misinformation vary across research fields according to falsity, misleadingness, and intentionality.

This distinction is important for automated detection because intention cannot be reliably established from textual content alone. Accordingly, the present study uses the broader classification of Misleading and Non-Misleading information rather than attempting to determine whether the author intentionally deceived the reader.

As misleading information increasingly appears in social-media posts, news content, and other user-generated text, automated detection has developed from

conventional lexical approaches toward contextual language models. Traditional methods commonly represent documents through statistical features such as

Term Frequency–Inverse Document Frequency, whereas contemporary approaches use pretrained transformers to capture relationships among words within context. Mouratidis et al. (2025) noted that both conventional machine- learning representations and contextual embeddings remain relevant in fake-news detection because each captures different properties of text. Consequently, evaluating a more complex representation against a conventional baseline is necessary before concluding that the additional complexity improves classification performance.

Among transformer architectures, DistilBERT provides a computationally reduced alternative to BERT. Sanh et al. (2019) introduced DistilBERT through knowledge distillation, reducing the size of BERT while retaining much of its language-representation capability. Although published before the principal 2020– 2026 range of this review, the work is retained because it is the foundational publication for the architecture employed in the present study. Its reduced computational requirements are particularly relevant to a system intended for interactive classification, where model performance must be considered together with practical inference requirements.

However, the effectiveness of contextual language models cannot be assumed to transfer uniformly to multilingual or code-switched text. Winata et al.

(2023) described code-switching as a continuing challenge in natural language processing because a single utterance may combine languages with different lexical, grammatical, and contextual properties. Multilingual language models can reduce some language-resource limitations, but their general multilingual ability does not automatically

guarantee equivalent performance on code-switched inputs. Therefore, a model intended for Taglish must be evaluated using actual Tagalog-English code-switched data rather than relying solely on results obtained from monolingual English or Filipino corpora.

This concern is especially significant in misleading-information detection, where language resources remain unevenly distributed. Low-resource languages frequently have fewer labeled datasets and specialized models than English, limiting both model development and reliable evaluation. Such limitations make locally appropriate dataset construction important because linguistic patterns learned from high-resource English news cannot simply be assumed to represent Filipino or Taglish online discourse. The present study consequently treats Taglish not merely as translated English text but as a code-switched linguistic setting requiring its own eligibility rules, representation, and evaluation.

Contextual representations alone may also overlook explicit characteristics of online writing. Garg and Sharma (2022) demonstrated that linguistic characteristics involving readability, lexical patterns, writing style, and psycholinguistic information can contribute to fake-news classification. At the same time, linguistic cues must be interpreted cautiously. Przybyla (2020) showed that a classifier can obtain apparently strong results by learning source- or topic-related characteristics rather than generalizable indicators. Thus, stylistic patterns should not be regarded as direct evidence that a statement is false; instead, they can be treated as supplementary numerical features whose contribution must be established through comparative evaluation.

This distinction is particularly important for informal code-switched communication. Taglish text may naturally contain Filipino particles, rapid alternation between English and Filipino, repeated characters, capitalization, unconventional punctuation, emotionally expressive terms, and urgency expressions. Such features may characterize the manner in which a message is written without independently determining its factual accuracy. For this reason, the present study examines these characteristics together with contextual text representations rather than converting them into fixed rules for identifying misleading information. The proposed hybrid design therefore rests on the principle that contextual embeddings and explicit linguistic features may contain complementary information, while their actual contribution must be demonstrated empirically.

The validity of any classification model nevertheless depends heavily on the quality of the data from which it learns. Gebru et al. (2021) emphasized that machine-learning datasets should be documented in terms of their motivation, composition, collection, preprocessing, intended use, and limitations. More recent analysis by Klie et al. (2024) further showed that erroneous labels, inadequate validation, and weak annotation-quality management remain present even in published NLP datasets. Together, these works indicate that simply increasing the quantity of samples does not ensure a reliable dataset; collection procedures, labeling criteria, annotator consistency, and validation must also be documented and controlled.

Dataset balance is another important consideration. In binary classification, substantial differences between class frequencies can cause a model to favor the more frequently represented class. Recent misinformation research has therefore used approximately balanced real and false classes to reduce majority-class effects and make

class-specific performance easier to interpret. However, balance should not be restricted to the overall number of labels. Topic composition must also be examined because a model can learn topic-label associations rather than the intended linguistic or contextual distinction. For example, when most false samples concern politics while most true samples concern entertainment, topic vocabulary itself may become an unintended shortcut for classification. Recent misinformation research demonstrates that maintaining both class balance and topical diversity can reduce such confounding effects.

Accordingly, the dataset of the present study is designed to distribute both Misleading and Non-Misleading samples across several information domains rather than concentrating either label within a particular subject. The precise Philippine-relevant categories are established further in the local literature; however, the broader methodological principle follows research showing that topic imbalance can distort apparent model performance. Equal or approximately equal representation is therefore used as a control mechanism, not because naturally occurring misinformation is assumed to have an equal real-world distribution.

Related to topic bias is the problem of source bias. Przybyla (2020) found that models trained on particular publishers or domains may learn source characteristics and then perform poorly when evaluated on unfamiliar sources.

This means that a classifier should not be allowed to associate a particular source automatically with one label. Dataset collection should therefore obtain evidence from multiple sources where possible and review whether publisher identity, domain, or recurring formatting has become disproportionately associated with either class.

A further threat to valid evaluation is data leakage. Kapoor and Narayanan (2023) identified leakage as a major source of exaggerated machine-learning performance because information related to the evaluation set can unintentionally become available during model development. In misleading-information datasets, this issue is especially important because one underlying claim may occur as different headlines, paraphrases, reposts, translations, or social-media versions. If closely related versions of the same claim appear in both training and testing data, the classifier may partly memorize that claim rather than demonstrate generalization to previously unseen information.

For this reason, the organization of training, validation, and testing data must consider semantic relationships among samples rather than only randomly dividing individual rows. Van der Goot (2021) likewise emphasized that dataset splits should reflect the research question and preserve the independence of the final test set. The training set is used for model learning, the validation set supports model and parameter selection, and the test set should remain unseen until final evaluation. These principles provide the basis for separating related claim groups and maintaining independent dataset partitions in the present study.

The quality of the ground-truth labels is equally important. Murayama et al. (2022) demonstrated that fake-news datasets can be constructed through explicit annotation questions and multiple independent annotators and evaluated annotator consistency through agreement statistics. Building on similar concerns, Klie et al. (2024) found that high-quality NLP dataset construction commonly involves clear instructions, annotator management, validation, adjudication of disagreements, and appropriate inter-annotator agreement measures rather than relying only on simple percentage agreement.

These principles are directly applicable to the present study because the distinction between Misleading and Non-Misleading information should be based on evidence and predefined annotation rules rather than the researchers' personal impressions. Annotation must therefore distinguish factual assessment from linguistic style, specify how ambiguous or unverifiable content is treated, require independent judgments before disagreement resolution, and measure consistency among annotators. This helps prevent subjective characteristics such as emotional wording, capitalization, or urgency from becoming the basis for the ground-truth label when those same characteristics are later evaluated as model features.

Reliable model assessment further requires multiple classification measures.

Accuracy summarizes the proportion of correct predictions but does not indicate how errors are distributed between classes. Consequently, misinformation- detection research commonly reports precision, recall, and F1-score together with a confusion matrix to distinguish true positives, true negatives, false positives, and false negatives. These measures are particularly important when the consequence of incorrectly flagging legitimate information differs from that of failing to identify misleading information. The present study therefore evaluates both overall classification performance and class-specific errors rather than relying on accuracy alone.

In addition, any numerical confidence displayed by a classifier must be interpreted as an output of the model rather than independent evidence that a statement is objectively true or false. Desai and Durrett (2020) showed that probability estimates produced by pretrained transformer classifiers can be poorly calibrated, particularly when the

characteristics of test data differ from those encountered during training. Thus, if the developed system presents a confidence percentage, its interpretation should remain tied to the model's prediction and should not be described as a factuality probability.

Taken together, indicates that a dependable misleading-information classifier requires more than selecting a high-performing language model. Its validity also depends on language-appropriate data, complementary but carefully interpreted linguistic features, documented annotation procedures, control of class, topic, and source bias, protection against data leakage, independent evaluation data, and appropriate performance measures. These principles establish the theoretical and methodological foundation for the Taglish dataset and hybrid DistilBERT–Logistic Regression approach examined in the present study.

Local Literature

The Philippine information environment provides a distinct context for misleading-information detection because online news consumption, social-media use, and multilingual communication are closely intertwined. Chua (2025), writing for the Reuters Institute’s Digital News Report, observed that social media remain a preferred source of news among Filipinos and that political disinformation intensified alongside growing political polarization. The 2024 report similarly noted that online and social media were among the most widely used news sources in the country (Chua, 2024). These conditions increase the likelihood that users encounter information through platform feeds, shared posts, short-form content, and other forms of user-generated communication rather than exclusively through formal news websites.

The prevalence of online information also makes the Philippine setting particularly relevant to research on misinformation and public decision-making.

Valderama and Ong (2021), for instance, found that patterns of social-media engagement were associated with Filipino university students' susceptibility to political disinformation. More recently, Deinla et al. (2023) examined 23,996 college students across the Philippines and showed that misinformation susceptibility and confidence in identifying misinformation were relevant to voting- related behavior. Taken together, these findings indicate that misleading information is not merely a technical classification problem; it also concerns how users interpret and interact with information encountered online. For this reason, a detection system intended for Philippine users should function as an assistive screening mechanism that encourages further verification rather than as a substitute for independent judgment or professional fact-checking.

The problem is especially visible in political and governance-related communication. Mendoza et al. (2023) examined almost 24,000 social-media posts associated with the Tallano gold narrative and demonstrated how a persistent false claim could spread through Philippine political networks before and during the 2022 elections. Similar patterns remain evident in contemporary fact- checking efforts. Tsek.ph, a collaborative initiative involving academe, media organizations, and civil-society groups, documented false, misleading, altered, and context-deficient claims during the 2025 Philippine elections. Its published fact checks covered claims concerning public officials, government actions, elections, criminal proceedings, public funds, and manipulated or misrepresented media.

Thus, political misinformation in the Philippines extends beyond completely fabricated statements and can also involve genuine material presented with an incorrect or incomplete context.

However, misleading information in the country is not confined to politics. The COVID-19 period demonstrated how inaccurate online information can also affect public health communication. Lacsa (2022) emphasized the role of social-media and digital platforms in the COVID-19 infodemic, while Batara et al. (2023) showed the importance of social-media communication in the dissemination of pandemic- related information by Philippine local governments. These findings illustrate that information reliability becomes important whenever online claims concern public decisions, health, safety, or government services. Accordingly, the present study should not construct its dataset around a single domain such as elections, since doing so could result in a classifier that learns the vocabulary of one subject rather than more general characteristics associated with misleading content.

This broader Philippine information environment provides the basis for organizing the proposed dataset across multiple public-information domains.

Political and governance claims are clearly represented in Philippine misinformation research and fact-checking archives, while health-related misinformation became particularly prominent during the pandemic. Tsek.ph fact checks further demonstrate that misleading claims may concern public funds, public safety, government institutions, prominent individuals, and current social issues. Rather than treating every emerging issue as a separate class, the present study groups frequently encountered content into broader topical categories so that the dataset remains manageable while

retaining subject diversity. The categories are therefore intended as dataset-stratification domains, not as additional prediction labels.

The planned domains include Politics, Governance, and Elections; Health and Medicine; Economy, Finance, and Employment; Public Safety, Disaster, and Security; Science, Technology, and Environment; Education and Social Issues; Entertainment and Public Figures; and Other or General Information. These eight categories consolidate related information domains commonly encountered in Philippine online discourse and fact-checking activities. Their purpose is methodological: both Misleading and Non-Misleading samples should appear across different subjects so that the classifier is less likely to associate a particular topic with a particular label. The exact number of eight is therefore an adaptation made for the present dataset rather than a claim that previous Philippine research prescribes eight universal misinformation categories.

Language presents an additional challenge because Philippine online communication frequently moves between English and Filipino rather than remaining monolingual. Herrera et al. (2022) described Tagalog-English code- switching as an increasingly relevant low-resource NLP setting and introduced the TweetTaglish dataset containing more than 20,000 Tagalog-English code- switched instances. Their work demonstrated not only the availability of naturally occurring Taglish text but also the need for explicit procedures to identify and label code-switching instead of assuming that text containing occasional Filipino or English terms constitutes meaningful Taglish.

This distinction is important for the present research because the dataset is intended specifically for Taglish misleading-information detection. A sample should therefore satisfy an operational language criterion before being retained for annotation.

Herrera et al. (2022) provide an empirical basis for identifying Tagalog- English code-switching as a linguistic phenomenon that can be systematically represented in a dataset. Their findings also support preserving naturally occurring Taglish rather than translating monolingual material simply to increase the number of samples, since translated text may not reproduce the lexical choices, switching behavior, and informal structures found in authentic online communication.

The need for authentic Filipino-language resources is reinforced by Montalan et al. (2025), who developed Batayan, an NLP benchmark containing both Tagalog and code-switched Taglish data. Their evaluation identified continuing performance gaps in models processing Filipino and emphasized native-speaker- driven adaptation and validation to reduce unnatural or translation-based artifacts.

Their findings complement Herrera et al. (2022): while TweetTaglish demonstrates the value of naturally occurring Tagalog-English code-switching data, Batayan further indicates that linguistic authenticity and native-speaker validation remain important even when modern multilingual language models are used.

Consequently, the present study should prioritize genuine Taglish samples and language-aware validation rather than treating Filipino-English data as interchangeable with translated English datasets.

These linguistic characteristics also affect preprocessing. Informal Taglish may contain abbreviations, particles, repeated characters, unconventional punctuation, capitalization, and rapid switching between Filipino and English. Completely normalizing such text may simplify input representation but can simultaneously remove characteristics that the study intends to examine as explicit linguistic or stylistic features. The dataset-

preparation procedure should therefore preserve the original version of each retained sample while applying only the preprocessing required by the model representation. In this way, code-switching and stylistic characteristics can be extracted from the original text without preventing the contextual model from receiving appropriately prepared input.

Local fact-checking practice provides an equally important basis for ground- truth validation. Tsek.ph does not limit questionable information to a simple true- versus-false distinction; its published assessments include classifications such as false, misleading, needs context, altered, and no basis. This reflects the reality that online claims may combine accurate and inaccurate elements or may become misleading because relevant context has been removed.

For a binary machine-learning study, these more detailed fact-checking judgments must therefore be mapped carefully rather than copied without rules.

Claims verified by credible fact-checking organizations as false, misleading, altered in a materially misleading manner, or unsupported may provide evidence for the Misleading class when they satisfy the study's annotation criteria.

Conversely, content should not automatically be assigned to the Non-Misleading class merely because no fact check criticizing it can be found. A non-misleading judgment also requires adequate supporting evidence. Ambiguous, purely opinion- based, satirical, unverifiable, or insufficiently documented content should be handled according to the annotation codebook rather than forced into either class.

This evidence-oriented approach is important because linguistic form alone cannot establish factuality. An emotionally worded or highly capitalized post may still contain accurate information, while professionally written content may contain a misleading claim.

The proposed annotation therefore separates ground-truth determination from feature extraction: factual labels are established through evidence and the annotation procedure, whereas code-switching, punctuation, capitalization, urgency, emotional language, and related characteristics are subsequently measured as model inputs. Maintaining this separation reduces the risk of circular reasoning in which the same stylistic characteristics used by the model are also used by annotators to decide the correct label.

The Philippine context also raises requirements concerning the responsible collection of online data. The National Privacy Commission's implementing rules for the Data Privacy Act permit the processing of publicly available personal data for research under specified circumstances but require appropriate safeguards and respect for data-subject rights. The rules further emphasize transparency regarding the purpose and nature of data processing. Therefore, collecting publicly accessible online text does not eliminate the researchers' responsibility to minimize unnecessary personal information, document the purpose of collection, and implement appropriate privacy protections.

For the proposed dataset, only information necessary for textual classification should consequently be retained. Usernames, profile identifiers, contact details, and other personally identifying information that do not contribute to the research objective should not form part of the model's training features. Source information necessary for evidence verification and dataset auditing may be recorded separately under controlled access rather than exposed through the classification model. This allows the dataset to remain traceable for research validation while minimizing unnecessary personal-data processing.

The same principle applies when the system receives text from different online formats. Philippine users frequently encounter information through ordinary webpage

text, social-media posts, images containing written claims, and videos containing either visible text or spoken information. These forms do not require the study to redefine the classifier as a multimodal model. Instead, Optical Character Recognition and speech-to-text may serve as input-extraction mechanisms that convert supported visual text or spoken content into the textual form required by the same classifier. The machine-learning decision therefore remains text-based regardless of whether the text was directly available, pasted by the user, extracted through OCR, or transcribed from speech.

This distinction keeps the system's capability aligned with the research problem. A video may contain manipulated imagery, facial expressions, synthetic voices, edited footage, or other non-textual cues, but these are outside the classifier's scope unless separate visual or acoustic models are developed.

Similarly, OCR identifies visible written characters but does not determine whether the surrounding image itself has been digitally manipulated. The proposed system consequently evaluates only the extracted or supplied textual content and should state this limitation consistently in the scope, system architecture, and user interface.

Overall, the local literature establishes three important conditions for the present study. First, misleading information in the Philippines occurs within an information environment where social media remain highly influential and misleading narratives extend across political, health, public, and social concerns.

Second, Tagalog-English code-switching requires linguistically authentic data and validation procedures appropriate to Filipino communication rather than direct dependence on English-language resources. Finally, Philippine fact-checking and privacy practices support evidence-based labeling and responsible treatment of collected online

material. These conditions provide the local foundation for developing a multi-topic, evidence-validated Taglish dataset and deploying its classifier as a text-based screening system for Philippine users.

Review of Related Studies

Foreign Studies

Recent studies on misleading-information detection show that model performance depends not only on the selected algorithm but also on how the dataset is constructed, divided, annotated, and evaluated. This is especially important in low-resource settings, where increasing the number of samples is often constrained by the cost of manual verification. Setiawan et al. (2025), for instance, examined fake-news detection under limited-data conditions by deliberately using a subset of 4,000 samples from the larger Fakeddit dataset. The authors selected this amount to simulate a setting in which labeled data are scarce and found that useful classification performance could still be obtained despite the reduced dataset. Although their work involved multimodal data rather than Taglish text, it provides an empirical precedent for using 4,000 samples as a manageable low-resource benchmark, particularly when manual validation is required. In the present study, this value is therefore adapted as the minimum dataset target rather than treated as a universally prescribed sample size.

The usefulness of a dataset, however, is determined not only by its size but also by how its samples are allocated for model development and evaluation. A recent bilingual fake-news corpus developed by Lamasheva et al. (2026) divided its data using

a stratified 70:15:15 ratio for training, validation, and testing while preserving both language and REAL/FAKE class proportions. A similar distribution was used in a multilingual fake-news study in which 70% of the data were assigned for training and 15% each for validation and testing, again through stratified sampling to maintain class balance. These studies provide a direct methodological basis for adopting a 70:15:15 partition in the present research. More importantly, the use of stratification ensures that the proportions of Misleading and Non- Misleading samples are retained across the three partitions instead of allowing one subset to contain an unintended concentration of either class.

For a dataset of 4,000 validated instances, this allocation corresponds to 2,800 training samples, 600 validation samples, and 600 testing samples. The training portion supplies the examples from which the classifier learns, while the validation portion is reserved for model and parameter selection. The test set, in contrast, remains separate until final evaluation. Van der Goot (2021) emphasized that development and test partitions serve different purposes and that repeated use of the test set during model development can undermine the validity of the reported results. Similarly, Kapoor and Narayanan (2023) identified leakage between training and evaluation data as a major cause of overly optimistic machine-learning performance. Thus, the proposed split is not justified solely by its numerical ratio; its validity also depends on keeping the testing data independent throughout development.

This separation becomes more difficult in misinformation datasets because different posts may communicate the same underlying claim using slightly different wording. Randomly splitting individual records can therefore place one version of a claim

in the training set and another paraphrased or reposted version in the test set. Under such conditions, a classifier may reproduce a claim it has already encountered rather than demonstrate its ability to handle previously unseen information. Kapoor and Narayanan (2023) showed that this form of non- independent sampling can inflate machine-learning performance, while van der Goot (2021) similarly argued that dataset splitting should reflect the structure of the research problem rather than relying mechanically on random partitions.

Following these principles, the present study groups duplicate, near-duplicate, paraphrased, translated, or reposted instances of the same underlying claim before dataset splitting so that one claim group is restricted to a single partition.

Before any division of the data, the reliability of the labels must first be established. Murayama et al. (2022), in constructing a Japanese fake-news dataset, developed a structured annotation scheme after examining existing fake- news datasets and used multiple annotators to evaluate different characteristics of questionable information. Their work demonstrates that fake-news labeling should be guided by explicit criteria rather than by an annotator's immediate impression of a text. Dzienisiewicz et al. (2024) applied a comparable approach in the development of the POLygraph dataset, which combined expert and non-expert manual annotation in constructing a large resource for fake-news detection.

Together, these studies support the use of written annotation guidelines and independent human judgments before the labels are accepted as ground truth.

The annotation procedure can be strengthened further through a preliminary annotation stage. Klie et al. (2024), after examining 591 publications introducing text

datasets, found that annotation quality is commonly managed through practices involving annotator instructions, agreement measurement, adjudication, and data validation. Their analysis also showed that improper use of agreement measures remains a recurring weakness in dataset development. These findings provide the basis for conducting a pilot annotation before the full dataset is labeled, allowing unclear definitions, overlapping categories, and inconsistent interpretations to be identified while the annotation rules can still be revised.

Following the pilot, the researchers may proceed with the established codebook rather than modifying the criteria after large portions of the dataset have already been labeled.

Because the proposed study involves several researchers assigning categorical judgments, agreement must also be quantified rather than inferred from discussion alone. Murayama et al. (2022) assessed agreement among annotators as part of their annotation process, while Klie et al. (2024) emphasized that the agreement statistic should correspond to the actual annotation design. For more than two annotators assigning categorical labels to the same set of items, Fleiss' kappa provides an appropriate measure of inter-annotator agreement.

Accordingly, if all four researchers independently label the same pilot records, Fleiss' kappa may be used to determine whether the annotation guidelines are being applied consistently. Disagreements can then be examined through an established adjudication procedure, while records that remain unsupported or irreconcilably ambiguous can be excluded rather than assigned a forced label.

This procedure is particularly important because stylistic characteristics should not determine the ground-truth label. A post containing emotional wording, repeated punctuation, capitalization, or urgency may still be factually correct.

Conversely, misleading information may be written in a neutral and professional form. Murayama et al. (2022) showed that fake news can involve several dimensions beyond simple surface wording, while Garg and Sharma (2022) demonstrated that linguistic characteristics can nevertheless provide useful discriminatory information when treated as model features. Hence, the annotation stage and feature-extraction stage must remain separate: the former establishes whether the information is supported by evidence, while the latter measures linguistic patterns that may contribute to classification.

Garg and Sharma (2022) provide a further empirical basis for combining conventional text representations with explicit linguistic information. Their framework evaluated 26 linguistic features alongside TF-IDF, Count Vectorizer, and Hash Vectorizer representations across four fake-news datasets. The results showed that linguistic characteristics can supplement conventional text representations in automatic fake-news detection. Rather than adopting all 26 features directly, the present study adapts the broader principle of combining textual representation with a smaller set of features suited to informal Taglish communication, including code-switching patterns, Filipino particles, punctuation intensity, capitalization, repeated characters, emotional-language cues, and urgency-related expressions. These features are therefore selected as measurable supplementary signals and not as independent rules for declaring a sample misleading.

The contextual component of the proposed model is likewise supported by studies that combined transformer-derived information with conventional machine- learning classification. Ayoub et al. (2021) compared several approaches for COVID-19 misinformation detection and used a DistilBERT–Logistic Regression configuration in which knowledge obtained from BERT-based representations was used with Logistic Regression. Their results showed competitive performance while retaining a simpler classification layer, demonstrating that contextual transformer representations can be combined with a conventional probabilistic classifier rather than requiring the transformer itself to serve as the final classifier.

This configuration is particularly relevant to the present study because it provides a precedent for using a lightweight contextual representation together with Logistic Regression.

Nevertheless, a hybrid model cannot be considered superior merely because it contains additional features. Its contribution must be measured against simpler configurations under the same evaluation conditions. Garg and Sharma (2022) compared multiple representations and classifiers rather than evaluating a single model in isolation, while recent fake-news studies similarly retain conventional baselines when testing transformer-based approaches. Following this practice, the proposed hybrid model should be evaluated against at least a conventional TF- IDF with Logistic Regression baseline and a contextual representation baseline.

Maintaining Logistic Regression as the final classifier where configurations permit makes it possible to determine whether observed differences are primarily associated with the representation being supplied to the classifier.

The same principle applies to the seven proposed linguistic features. Their contribution should be supported by comparative performance rather than assumed in advance. If the contextual representation alone performs similarly to or better than the fused representation, the additional linguistic branch cannot be claimed to improve detection. Conversely, an improvement in validation and held- out test performance would provide empirical support for retaining the hybrid design. In this way, model selection is based on measured results rather than on the assumption that a more complex configuration necessarily performs better.

Class distribution must also be controlled during these experiments. Balanced or stratified datasets are frequently used in fake-news research to prevent a model from obtaining high overall accuracy primarily by predicting the majority class. The bilingual corpus of Lamasheva et al. (2026), for example, preserved both class and language proportions during stratified partitioning, while the multilingual study reported in 2025 likewise applied stratified sampling to maintain class balance.

Adapting this principle, the present dataset is planned to contain equal numbers of Misleading and Non-Misleading samples. For the 4,000-sample target, this corresponds to approximately 2,000 instances per class. The balanced composition is therefore a controlled research design for model comparison and should not be interpreted as evidence that misleading and non-misleading information occur equally often in actual online environments.

Beyond overall class balance, stratification across topics is needed to reduce the possibility that topic vocabulary becomes a substitute for the intended classification task. Thus, when the dataset is distributed across the eight domains established in the review

of Philippine literature, both labels should be represented within each domain as far as the available validated data permit. This creates approximately 500 samples per topic under an equal eight-topic distribution, with approximately 250 Misleading and 250 Non-Misleading instances in each category. The values serve as target quotas for controlling dataset composition rather than assumptions about the natural prevalence of misinformation. The final distribution should consequently be reported transparently, particularly when evidence availability prevents an exact quota from being reached.

After training, the selected models must be evaluated with measures that distinguish different kinds of classification performance. Setiawan et al. (2025) assessed limited-data fake-news models using measures including accuracy, precision, and recall, while multilingual fake-news experiments have similarly reported F1-based measures in addition to overall accuracy. These metrics address different aspects of prediction: accuracy represents the overall proportion of correct classifications, precision reflects the proportion of predicted positive cases that are correct, and recall measures the proportion of actual positive cases successfully detected. F1-score combines precision and recall and is useful when both false positives and false negatives are important. The confusion matrix complements these measures by displaying the numbers of correct and incorrect predictions for each class.

For the present study, the false-positive rate should also be reported because labeling legitimate information as potentially misleading has a direct effect on the usefulness of the screening system. A model can achieve acceptable overall accuracy while still producing an undesirable number of false alarms. Therefore, model selection

should consider the pattern of errors together with accuracy, precision, recall, and F1-score instead of choosing a configuration from a single metric.

The system may also display a numerical confidence associated with a classification result. However, confidence values derived from classifier probabilities are not necessarily equivalent to the observed probability that a prediction is correct. Desai and Durrett (2020) showed that pretrained transformer classifiers can be miscalibrated and that calibration should be examined when prediction probabilities are interpreted by users. Accordingly, the system's percentage output should represent the model's classification confidence rather than a percentage of factual truth. Where the final implementation displays such a score, calibration analysis should be performed before the value is treated as a meaningful probability.

Although the classifier itself remains text-based, some supported online information may first require text extraction. Optical Character Recognition provides one means of converting words embedded in images into machine- readable text. Nazeem et al. (2024) compared several open-source OCR libraries, including Tesseract, EasyOCR, MMOCR, PaddleOCR, and Keras OCR, and showed that OCR systems differ in accuracy across languages and inputs.

Similarly, later benchmarking of seven OCR engines found measurable differences in accuracy, Character Error Rate, Word Error Rate, and execution time. These results indicate that OCR should be regarded as a separate extraction stage whose errors may influence the subsequent text classification. If EasyOCR is retained in the system, its selection should therefore be based on implementation compatibility and

empirical testing on representative input rather than on an assumption that it is universally the most accurate OCR engine.

A comparable limitation applies to speech-to-text processing. Ugan et al. (2024) found that code-switched speech continues to present challenges for automatic speech recognition systems, while Aditya et al. (2024) specifically noted that multilingual models such as Whisper can struggle when languages are mixed within the same utterance. Their findings support the use of Whisper as a multilingual transcription tool while also indicating that transcription accuracy must not be assumed for Taglish speech. Hence, when spoken content from supported videos is transcribed before classification, the resulting transcript should be treated as extracted text, and transcription errors should be acknowledged as a possible source of downstream classification error.

Altogether, the reviewed foreign studies provide empirical precedents for the principal methodological procedures of the study. The 4,000-sample target is adapted from limited-data fake-news experimentation by Setiawan et al. (2025), while the stratified 70:15:15 partition is adapted from recent multilingual and bilingual fake-news experiments. Dataset labeling is adapted from structured fake- news annotation practices demonstrated by Murayama et al. (2022) and Dzienisiewicz et al. (2024) and is strengthened by the annotation-quality recommendations of Klie et al. (2024). Meanwhile, Garg and Sharma (2022) support the incorporation of explicit linguistic features, and Ayoub et al. (2021) provide a precedent for combining DistilBERT-derived contextual information with Logistic Regression. These procedures are not copied without modification; rather, they are adapted to the linguistic characteristics, evidence

requirements, topic distribution, and intended deployment of the Taglish dataset developed in the present research.

Local Studies

Studies conducted in the Philippine setting demonstrate that automated misleading-information detection faces two closely related limitations: the scarcity of validated Filipino misinformation data and the difficulty of modeling informal, multilingual language. Cruz et al. (2020) addressed the first limitation by developing the Fake News Filipino dataset, an expertly curated corpus of 3,206 Filipino news samples with an equal distribution of real and fake articles. Using transfer-learning techniques, the researchers achieved 91% accuracy and further improved performance to 96% through auxiliary language-modeling objectives.

Their findings showed that useful classification performance can be achieved even with relatively limited Filipino-language data when the dataset is carefully constructed and appropriate transfer-learning methods are used. At the same time, the dataset consisted primarily of Filipino news articles rather than naturally occurring Taglish posts, leaving code-switched misleading-information detection less explored.

The balanced construction of Fake News Filipino also provides a local precedent for controlling class distribution. Cruz et al. (2020) used an equal number of real and fake samples rather than allowing one class to dominate the dataset. This approach is consistent with the foreign studies discussed earlier, in which stratification and class balance were used to reduce majority-class effects during model development. Accordingly, the planned equal distribution of Misleading and Non-Misleading samples in the present dataset follows both local and foreign empirical practices. However, the

balance is maintained for controlled model development and comparison; it does not imply that misleading and non- misleading information naturally occur at equal rates online.

Cruz et al. (2020) further evaluated their models across political news, entertainment news, and opinion articles and reported that the transfer-learning approach generalized across these content types. This finding strengthens the need to consider topic variation when constructing a Philippine misinformation dataset. Nevertheless, the range of subjects in their corpus does not cover the broader set of public-information domains included in the present research.

Consequently, the current dataset extends topical coverage beyond politics and entertainment while maintaining the same underlying principle that model performance should not depend on a single subject area. The eight-topic organization developed for the present study therefore expands, rather than directly reproduces, the topical organization of earlier Filipino fake-news research.

Although Fake News Filipino established an important local benchmark, its linguistic composition does not fully represent the code-switching commonly observed in Philippine online communication. Herrera et al. (2022) addressed this gap from a language-resource perspective through TweetTaglish, a corpus containing more than 20,000 Tagalog-English code-switched instances collected from Twitter. Instead of treating every mixture of Filipino and English as equivalent, the researchers annotated language use at the token level and developed measures for examining the degree of code-switching within individual tweets.

Their results showed that Tagalog-English code-switching can be systematically identified and quantified, providing a stronger basis for determining whether candidate samples genuinely contain both linguistic components.

However, TweetTaglish was created to investigate code-switching rather than misinformation. Conversely, Fake News Filipino provides misinformation labels but does not specifically address naturally occurring Taglish. Considered together, the two datasets reveal a methodological gap: one provides the classification task, while the other provides the linguistic setting required by the present research.

Thus, instead of assuming that an existing Filipino fake-news dataset is automatically suitable for Taglish classification, candidate records must first satisfy a defined Taglish eligibility criterion before undergoing evidence-based annotation.

This also prevents monolingual Filipino or English records from being retained merely to meet the required dataset size.

The importance of preserving authentic Filipino and Taglish usage is reinforced by Montalan et al. (2025), who developed Batayan, a Filipino NLP benchmark covering eight tasks across understanding, reasoning, and generation. The benchmark contains both Tagalog and code-switched Taglish data and was developed through native-speaker-driven adaptation and validation. Their analysis showed that Filipino remains underrepresented in language-model pretraining and that direct translation can introduce unnatural vocabulary, syntax, and other forms of translationese. Furthermore, the evaluated language models continued to exhibit performance gaps on Filipino tasks despite their general multilingual capabilities.

The findings of Herrera et al. (2022) and Montalan et al. (2025) converge on the importance of linguistic authenticity. Herrera et al. demonstrated that naturally occurring Taglish can be systematically represented, whereas Montalan et al. showed that native-speaker validation is necessary to preserve natural Filipino usage and reduce translation artifacts. Therefore, the dataset should prioritize naturally occurring Taglish instead of generating a large portion of its samples by translating English or Filipino misinformation. Translation may increase dataset quantity, but it can also introduce language patterns that differ from those encountered by users in actual Taglish communication.

Local research also supports the use of contextual language models for Filipino fake-news detection. Espiritu et al. (2024) developed an improved DistilBERT model for Filipino fake-news classification and compared its performance with BERT using accuracy, precision, recall, F1-score, inference time, memory requirements, and model size. Although BERT obtained slightly stronger classification performance, DistilBERT required fewer computational resources and provided faster inference. Their findings demonstrate that model selection for Filipino misinformation detection should consider computational efficiency together with predictive performance, particularly when the classifier is intended for deployment rather than experimental evaluation alone.

This comparison is particularly important for the proposed browser-based implementation. A larger model may produce marginally stronger predictive results while requiring greater memory and inference resources, whereas a compressed transformer may provide a more practical balance between classification performance and computational demand. Espiritu et al. (2024) therefore provide a direct Philippine empirical basis for examining DistilBERT in the proposed model.

Their work also supports the use of accuracy, precision, recall, F1-score, inference time, and resource requirements when comparing model configurations rather than evaluating the classifier through accuracy alone.

Nevertheless, the study of Espiritu et al. (2024) focused on Filipino-language fake-news classification and did not examine the combination of contextual representations with explicit Taglish linguistic and stylistic features. This distinction becomes important when considered alongside Filipino code-switching research.

Taglish does not merely consist of Filipino text with occasional English vocabulary; its switching behavior, informal spelling, particles, and mixed-language constructions introduce characteristics that may not be represented adequately by a model developed primarily for monolingual Filipino data. Hence, the findings of Espiritu et al. (2024), Herrera et al. (2022), and Montalan et al. (2025) collectively suggest that an efficient contextual model is useful, but its performance must still be tested on authentic code-switched data rather than inferred from Filipino-only results.

The value of explicitly modeling code-mixed linguistic information has also been examined in Philippine text-classification research outside fake-news detection. A 2023 IEEE study on cross-lingual sentiment analysis developed a large-scale domain-specific Taglish corpus from higher-education feedback and examined cross-lingual word representations for Filipino-English code-mixed text. Although sentiment analysis differs from misleading-information detection, the study demonstrated that Taglish can be treated as a distinct computational setting in which information from both languages must be represented within the same classification task. This complements Herrera et al. (2022), whose work focused on identifying and measuring code-switching, and further

supports evaluating language-aware representations rather than treating Taglish as ordinary English text containing Filipino words.

Taking everything into account, these studies also explain why preprocessing must be conservative. Herrera et al. (2022) relied on token-level linguistic information to characterize Tagalog-English switching, while Montalan et al. (2025) emphasized natural Filipino vocabulary, sentence structure, and colloquial usage during dataset development. Removing or aggressively standardizing these characteristics before they are analyzed could eliminate information associated with actual Taglish usage. Consequently, the original form of each sample should be retained for linguistic-feature extraction, while model-specific preprocessing can be applied separately to the representation used by the contextual branch.

The reviewed local studies further indicate that dataset validity cannot be reduced to dataset size. Fake News Filipino contains 3,206 expertly labeled samples and nevertheless achieved strong transfer-learning results, whereas TweetTaglish contains more than 20,000 instances because it was designed for a different linguistic task. Batayan, meanwhile, placed particular emphasis on native- speaker adaptation and validation rather than maximizing corpus size alone.

These differences show that the number of records required depends on the task, labeling requirements, and quality of the available data. For the present research, the 4,000-sample target adopted from the foreign limited-data precedent should therefore be interpreted together with the Philippine evidence that carefully validated low-resource datasets can remain useful even when they are substantially smaller than large English-language corpora.

Moreover, these local datasets differ in what their labels represent. Fake News Filipino provides binary real and fake labels, whereas TweetTaglish identifies linguistic code-switching rather than factual reliability. Batayan likewise evaluates broader Filipino NLP competencies rather than misinformation. As a result, none of these resources can simply be relabeled as a complete Taglish misleading- information dataset without additional validation. Existing records may serve as baseline, supplementary, or candidate data only when their original labeling procedure is compatible with the operational definitions of the present study.

Otherwise, they must undergo the same eligibility and evidence-validation procedures applied to newly collected samples.

This distinction is particularly important if Fake News Filipino is used as a baseline source. Cruz et al. (2020) established its labels through expert curation, making the corpus valuable for Filipino fake-news research. However, its records should not automatically be counted toward the final Taglish dataset because the language requirement of the present study is narrower. Only records that satisfy the operational Taglish criterion should be eligible for inclusion in the Taglish corpus; records that do not meet the criterion may still be used for preliminary comparison or supplementary experimentation, provided that their role is reported separately.

The same separation should be maintained between language validation and factual validation. Herrera et al. (2022) provide procedures for examining whether a sample exhibits Tagalog-English code-switching, while the fake-news dataset of Cruz et al. (2020) demonstrates the importance of expert-supported factual labels.

Montalan et al. (2025), in turn, emphasize native-speaker review for linguistic authenticity. These procedures address different dimensions of dataset quality and should not be collapsed into a single decision. A record may be authentic Taglish but factually unverifiable, just as a well-supported factual statement may fail the Taglish eligibility criterion. The final dataset must therefore satisfy both requirements before a sample is retained.

Local studies consequently establish a progression in Philippine NLP research. Cruz et al. (2020) demonstrated that a carefully curated Filipino fake-news dataset can support effective transfer learning despite limited resources; Herrera et al.

(2022) established a substantial corpus and systematic representation of naturally occurring Tagalog-English code-switching; Espiritu et al. (2024) showed that DistilBERT can provide a computationally efficient alternative for Filipino fake- news classification; and Montalan et al. (2025) demonstrated the continuing need for linguistically authentic, native-speaker-validated Filipino and Taglish resources.

When considered together, however, these works leave the intersection of the four concerns less developed: evidence-validated misleading-information labels, authentic Taglish input, contextual and explicit linguistic representations, and deployment through an accessible text-screening system. This intersection forms the empirical gap addressed by the present study.

Synthesis of Related Works

The reviewed foreign and local literature and studies indicate that misleading-information detection requires the integration of appropriate language representation, reliable dataset construction, and controlled model evaluation.

Transformer-based models such as DistilBERT provide contextual representations of text, while linguistic and stylistic features can supply additional information that may not be fully represented by contextual embeddings alone (Garg & Sharma, 2022; Espiritu et al., 2024). However, studies on code-switched language show that performance on monolingual data cannot simply be generalized to Taglish.

Herrera et al. (2022) and Montalan et al. (2025) emphasize the importance of authentic Tagalog-English data, appropriate language identification, and native- speaker validation. These findings support the use of naturally occurring Taglish samples and the examination of contextual and explicit linguistic features within the same classification framework.

Alongside model representation, the reviewed works consistently emphasize the importance of dataset quality. Cruz et al. (2020) demonstrated that a carefully curated and balanced Filipino fake-news dataset can support effective classification despite limited resources, while Setiawan et al. (2025) provided an empirical precedent for conducting fake-news detection using 4,000 samples in a limited-data setting. Accordingly, the present study adopts 4,000 validated Taglish samples as its minimum target, with equal representation of Misleading and Non- Misleading instances. Since class balance alone does not prevent a model from learning unintended topic or source patterns, the samples are further distributed across eight broad information domains derived from the Philippine information environment. The eight categories serve as dataset-stratification controls rather than additional prediction classes.

Moreover, reliable ground-truth labels require procedures that minimize subjective judgment. Murayama et al. (2022) and Klie et al. (2024) support the use of structured

annotation guidelines, independent judgments, pilot annotation, agreement measurement, and disagreement resolution in constructing labeled datasets. These principles provide the basis for the study's evidence-based annotation procedure and inter-annotator agreement assessment. Similarly, van der Goot (2021) and Kapoor and Narayanan (2023) emphasize maintaining independent dataset partitions and preventing data leakage. In response, the study adopts a stratified 70:15:15 training, validation, and testing split based on recent fake-news classification practices, together with claim-group separation to prevent duplicate, paraphrased, translated, or reposted versions of the same underlying claim from crossing dataset partitions.

The reviewed studies further suggest that a proposed hybrid model should not be assumed to outperform simpler approaches solely because it combines more features. Consequently, the hybrid DistilBERT–Logistic Regression configuration is compared with baseline representations to determine whether contextual and explicit Taglish linguistic features provide measurable improvement. Model performance is assessed through accuracy, precision, recall, F1-score, confusion matrix, and false-positive behavior, while any displayed confidence value is interpreted as model confidence rather than a direct measure of factual truth (Desai & Durrett, 2020). OCR and speech-to-text are likewise treated separately from the classification model, serving only to convert visible or spoken information into text before it enters the same text-based classification process.

Despite advances in Filipino fake-news detection and Tagalog-English NLP, a gap remains between these areas. Existing Filipino fake-news resources primarily address Filipino news content, whereas Taglish datasets mainly focus on code- switching and

other language-processing tasks rather than evidence-validated misleading-information classification (Cruz et al., 2020; Herrera et al., 2022).

Furthermore, the reviewed studies have not sufficiently examined the combination of contextual transformer representations and explicit Taglish linguistic features under an evidence-validated, multi-topic, and leakage-controlled dataset. The present study addresses this gap by developing a validated Taglish dataset and evaluating a hybrid DistilBERT–Logistic Regression model for misleading- information detection within a text-based browser system intended to support the screening of online information.

CHAPTER 3

METHODOLOGY AND DESIGN

This chapter presents the methodology and design used to develop and evaluate SuriPH. It covers the technical background, approaches and techniques, Agile Scrum-based iterative development, requirement analysis, respondents and sampling, research instruments, data gathering, statistical treatment, and the hardware, software, peopleware, network, and dataware requirements of the prototype.

Technical Background

The Armed Forces of the Philippines Information Technology Department at Camp Aguinaldo, Quezon City will serve as the institutional beneficiary and primary technical evaluator of this study. Qualified and authorized personnel can provide a professional assessment of the implemented academic prototype in areas such as functional suitability, performance efficiency, compatibility, interaction capability, reliability, security, maintainability, and flexibility where those characteristics are applicable and observable.

The department's participation will remain within an approved academic evaluation. SuriPH will not connect to AFP classified networks, intelligence systems, restricted communications, private institutional databases, or operational infrastructure. Any demonstration, temporary administrative access, or turnover will use only the approved prototype, approved test materials, and the environment authorized for the study.

The technical instrument will draw from selected applicable characteristics of ISO/IEC 25010:2023 (International Organization for Standardization, 2023).

The standard is a product-quality model, not a ready-made questionnaire or a certification awarded by this study. Each item must refer to implemented, observable behavior and pass adviser or expert review before use.

The intended public-facing users remain Filipino internet users who encounter English, Filipino, or Taglish online content. For formal general-user evaluation, the accessible population will consist of eligible fourth-year BSIT students from the selected class block who are at least 18 years old and can interact with online content in at least one of those languages. Their assessment will remain separate from the AFP IT technical evaluation.

SuriPH consists of a React-based browser extension, a restricted Admin Dashboard, a Python and FastAPI backend, media-to-text processing, machine- learning inference, independent credible-source retrieval, structured data storage, file and model storage, and controlled administrative maintenance. The following approaches describe how these components support the research objectives without extending the prototype beyond its approved academic scope.

Approaches and Techniques

Supervised Binary Text Classification

The study will use supervised binary text classification. Records accepted into an approved dataset version will carry the final study label Misleading or Non- Misleading. Records with inadequate context, insufficient evidence, unresolved disagreement, or non-checkable content may be excluded during dataset preparation. Exclude is a management status and not a prediction class.

Binary classification requires a clear relationship between the operational labels and the evidence used during validation. A record assigned Misleading must contain a checkable claim that conflicts with the approved evidence criteria.

A record assigned Non-Misleading must satisfy the approved criteria for the negative class. The procedure does not transform lack of evidence into proof that a claim is misleading, and it does not infer deceptive intent.

Records that cannot support a defensible binary decision will remain outside the approved model dataset. Exclude may be used for non-checkable opinion, missing context, insufficient evidence, duplicate material, unsupported language, unusable extraction, or unresolved reviewer disagreement. Revise may be used when the record can be corrected, segmented, or supplemented before another review. These statuses belong to dataset management and are not additional prediction classes.

Dataset Preparation and Evidence Validation

The Google Fact Check Tools API may assist reviewers by searching published fact-checked claims. A returned result will not assign the SuriPH label automatically. The reviewer must determine that the review addresses the same underlying claim and must record the evidence used (Google for Developers, 2024).

The approved dataset will be divided approximately into 70 percent training, 15 percent validation, and 15 percent testing data. Class stratification will be applied where compatible with claim grouping. The same partitions will be used for every experimental model configuration.

Candidate acquisition will begin with a source inventory identifying the resource, original purpose, content unit, language coverage, original label meaning, collection period where available, access conditions, and intended use.

Documented candidates include Fake News Filipino, BalitaNLP, NewsPH, the Philippine Fake News Corpus referred to by Jamen and Reyes (2023), and Cruz's (2024) Facebook-comment collection. The exact release and provenance of each imported resource must be confirmed. Formal news and social-media records will remain distinguishable for domain review. Public availability alone will not be treated as permission for unrestricted reuse, and unnecessary personal identifiers will be removed or protected.

Candidates will be screened for language, checkability, contextual completeness, text quality, provenance, topic, duplication, and evidence availability. Reviewers will check whether the evidence concerns the same date, subject, location, and factual proposition. Google Fact Check Tools API results may locate published reviews, while approved credible or authoritative sources may support cases without a matching fact check. Evidence publisher, rating where applicable, URL, claim-match decision, independent reviewer judgment, validation decision, notes, and final resolved label will be recorded. A multi-claim record may be revised, segmented, or excluded under the annotation guide when one label would conceal conflicting evidence.

Exact and near duplicates will be removed or reviewed before partitioning.

Text comparison and human claim review will identify substantially equivalent statements, which will share claim_group_id and remain in a single split. Source,

publisher, topic, language, and provenance metadata will support validation, leakage checks, and subgroup analysis but will not be direct classifier features.

These controls reduce leakage without proving that all source cues embedded in the text have disappeared.

The collection target is at least 4,000 evidence-annotated Taglish samples: approximately 2,000 Misleading and 2,000 Non-Misleading records distributed across eight topics. The team will begin with approximately 5,500-6,000 candidates to allow exclusions. The table states collection targets rather than guaranteed final counts; the achieved class, topic, source, exclusion, and claim-group distributions will be reported transparently after validation. Pure English and pure Filipino records will not be included in the primary experimental dataset.

Table 3.1 Dataset Collection Targets by Topic

Topic	Misleading Target	Non-Misleading Target	Total Target
Politics, Governance, and Elections	250	250	500
Health and Medicine	250	250	500
Economy, Finance, and Employment	250	250	500

Topic	Misleading Target	Non-Misleading Target	Total Target
Public Safety, Disaster, and Security	250	250	500
Science, Technology, and Environment	250	250	500
Education and Social Issues	250	250	500
Entertainment and Public Figures	250	250	500
Other and General	250	250	500
Total	2000	2000	4000

claimed before validation because evidence availability and exclusion decisions may change the usable distribution.

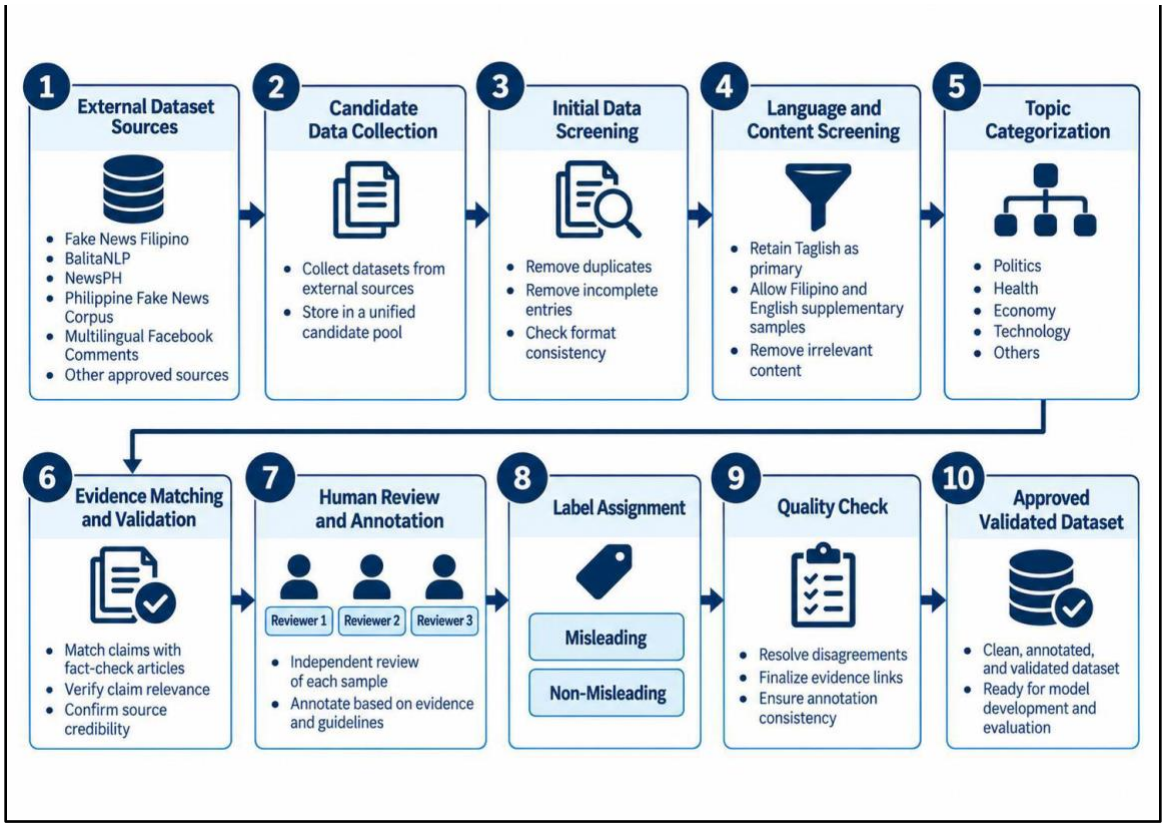


Figure 3.1 Dataset Screening and Evidence Validation Flow

All four researchers will independently annotate the same pilot subset of 100-150 screened records using the approved codebook and evidence form. Fleiss' Kappa will be calculated from their unreconciled binary label judgments. The obtained value will be reported with its interpretation; a value of at least 0.61 will be treated as substantial agreement for proceeding, not as proof of factual validity. If agreement is below this level or recurring disagreements reveal unclear rules, the team will revise the codebook, conduct calibration, and repeat an overlapping pilot before full annotation. During full annotation, each candidate will receive a primary annotation and an independent secondary verification. Disagreements will be discussed by the four researchers only after independent judgments are preserved. A record will be approved when its claim, context, and evidence support one label; it will be revised when segmentation or clarification

resolves the issue; and it will be excluded when evidence is insufficient, the claim is non-checkable, or disagreement remains unresolved.

Text Acquisition and Extraction

Browser content scripts will obtain selected supported social-media post content when extension permissions and URL-matching conditions allow access.

They will inspect the webpage DOM and send the resulting text or approved media-processing request to the backend. The backend will not inspect the DOM directly (Mozilla, n.d.-a).

Regular DOM and accessible or open Shadow DOM may be inspected where permitted. SuriPH will not claim access to closed shadow roots or every visible page structure. Shadow-root mode, browser and extension permissions, cross-origin restrictions, protected media, and webpage changes determine what can be obtained (Mozilla, n.d.-b).

The primary live inputs are user-selected text or captions from supported social-media post elements and text entered manually in the extension. EasyOCR and Whisper may provide optional text acquisition from supported images, screenshots, or bounded video segments when technically accessible. Their outputs remain unverified input text and are not treated as factual evidence.

For an explicitly requested and technically supported video segment, Whisper may transcribe accessible speech and EasyOCR may obtain visible text from sampled frames. Unsupported or protected media will return a clear status instead of a prediction from incomplete text.

Whisper was developed using large-scale multilingual and multitask speech supervision (Radford et al., 2023). The transcript and relevant frame text will be combined as textual input.

The request will identify selected supported post content rather than the entire feed or page. Compatibility testing will record the actual browser and extension version, platform or page type, permissions, supported structures, input type, expected behavior, and observed outcome. A successful test on one page structure will not establish support for every page on the same platform.

For images and screenshots, the backend will receive the supported file or captured region according to the implemented workflow. EasyOCR will detect text regions, recognized strings, and confidence information. Post-processing may order segments, remove duplicated detections, and discard unusable output. The original image will not be treated as a visual class input, and temporary media will be removed after the approved processing or testing period.

Video processing will obtain accessible audio and sample frames according to the implemented duration and resource limits. Whisper will transcribe speech, while EasyOCR will process visible text in selected frames. Repeated subtitles or captions may appear across several frames, so the media pipeline will deduplicate substantially identical segments before joining them with the transcript. The resulting text will enter the same preparation and classification pipeline as other inputs.

Acquisition errors will be recorded separately from classifier errors. Feasibility will be established through representative functional test cases with manually verified expected text, including successful extraction, unreadable input, unavailable audio,

protected media, timeout, and empty-output cases. CER and WER will not be separate study metrics; the study will report pass/fail outcomes and observed limitations so that media acquisition does not become a second full experiment or get conflated with classification performance.

Text Preprocessing

Text preparation may require encoding repair, whitespace normalization, removal of unusable extraction artifacts, and deduplication of repeated OCR segments. Punctuation, capitalization, repeated characters, digits, URLs, mentions, and hashtags will be preserved until the features that need them are computed. The final sequence and text version used by each branch will be recorded after implementation, rather than assumed in advance.

The contextual branch and TF-IDF baseline may require different transformations. The transformer tokenizer will apply its own subword segmentation and model-specific input construction. The TF-IDF configuration will build its vocabulary from training data only. The linguistic branch will compute features from a defined version of the prepared text. Each configuration will document the exact sequence so that validation and test records undergo the same fitted operations without learning from held-out data.

Input length is another practical constraint. Transformer models have a maximum sequence length, while webpage text or transcripts may exceed it. The implemented system will specify whether long text is truncated, segmented, pooled across chunks, or handled through another approved method. This decision will be evaluated because it affects both response time and the amount of claim context available to the classifier.

Contextual Feature Extraction

The contextual branch will use the distilbert-base-multilingual-cased checkpoint and its corresponding tokenizer. The model will generate a contextual representation from the prepared input sequence.

A documented pooling procedure will convert contextual hidden states into the classifier's input representation. Frozen versus fine-tuned use, pooling, long- text handling, and final dimensionality remain implementation-dependent. The checkpoint's hidden size alone does not establish the dimensionality of the final representation (Hugging Face, n.d.).

The contextual branch may be used as a frozen representation generator or may undergo controlled training depending on the final experiment. The decision affects computing cost, risk of overfitting, and reproducibility.

Hyperparameters, random seeds, model checkpoints, tokenizer version, pooling method, and any fine-tuning procedure will be recorded in the model-evaluation record.

All contextual experiments will use the same approved partitions. If representations are cached to reduce repeated computation, the cache will remain linked to the dataset version and preprocessing configuration. This prevents embeddings produced from one version of the data from being combined silently with another version during later experiments.

Linguistic and Stylistic Feature Extraction

The second branch will compute exactly seven explicit linguistic and stylistic features from the same preprocessed record: (1) code-switching ratio, calculated as the smaller of the Filipino-token and English-token counts divided by the total number of

Filipino and English tokens; (2) Filipino-particle frequency, calculated as matched tokens from the documented Filipino-particle lexicon divided by total tokens; (3) punctuation intensity, calculated as exclamation and question marks divided by total characters; (4) capitalization ratio, calculated as uppercase alphabetic characters divided by all alphabetic characters; (5) repeated-character ratio, calculated as characters belonging to runs of three or more identical characters divided by total characters; (6) emotional-language frequency, calculated as matches from the documented emotion lexicon divided by total tokens; and (7) urgency-cue frequency, calculated as matches from the documented urgency lexicon divided by total tokens. Zero-denominator cases will return zero. Lexicons, tokenization rules, normalization, and source code versions will be archived with the experiment.

Each feature is a measurable textual indicator rather than proof of misleadingness. All seven values will be computed before destructive normalization, scaled using parameters fitted only on the training set when scaling is required, and reused unchanged for validation, testing, and live screening. Their contribution will be tested through the linguistic-only and hybrid configurations.

Counts and ratios can have different numerical scales. Any required scaling will be fitted only on training data and reused unchanged for validation, testing, and live screening. The feature inventory will document each implemented definition, denominator, text-normalization stage, and handling of empty input. Sentiment, emotion, urgency, readability, code-switching scores, and TF-IDF or N-gram indicators will not be implied as hybrid features merely because related studies use them.

The linguistic-only experiment will establish whether the explicit measurements carry useful signal without contextual embeddings. It also provides a direct way to inspect coefficients and error patterns. The hybrid experiment will show whether those measurements add value after contextual representation is already present. If the hybrid does not improve held-out performance or increases false positives, the study will report that result rather than assume the added branch is beneficial.

Feature Fusion and Logistic Regression

The fused vector will enter a regularized Logistic Regression classifier implemented through scikit-learn. Logistic Regression supports dense and sparse input and can return class probability estimates (scikit-learn developers, n.d.).

Feature fusion requires the contextual and explicit vectors to be aligned by record and converted into compatible numerical structures. The process will verify that every row in the contextual matrix corresponds to the same approved sample as the linguistic row. Missing or failed representations will be handled through an approved rule rather than silently shifting row order.

Logistic Regression settings, including solver, regularization strength, class weighting, iteration limit, and decision threshold, will be selected using training and validation data and then recorded with the final configuration. No value is assumed here. The threshold and probability mapping will be frozen before held-out testing, and the same mapping will be used for the reported model and its public labels.

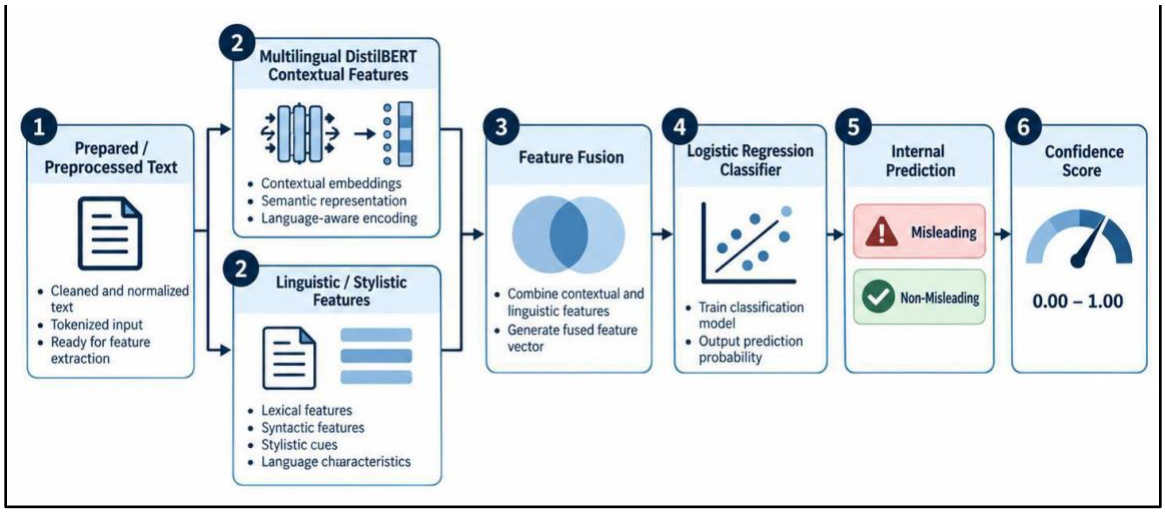


Figure 3.2 Hybrid DistilBERT-Logistic Regression Model Pipeline

Baseline Model Comparison 1. TF-IDF with Logistic Regression 2. Multilingual DistilBERT contextual representation with Logistic Regression 3. Linguistic and stylistic features with Logistic Regression 4. Proposed hybrid multilingual DistilBERT and linguistic-feature representation with Logistic Regression The four configurations answer different questions. TF-IDF with Logistic Regression measures the strength of a conventional lexical baseline. Multilingual DistilBERT representations with Logistic Regression measure contextual information without explicit features. Linguistic and stylistic features with Logistic Regression measure the selected explicit branch. The hybrid configuration measures whether fusion provides additional value over these controlled alternatives.

Macro F1 will be the primary overall measure because it gives equal weight to the class-level F1 values and reduces the ability of a larger class to dominate the summary. Accuracy, precision, recall, false positive rate, class-level scores, and confusion matrices will provide complementary interpretation. Taglish results and other required subgroups

will be reported separately when the available sample size supports a meaningful comparison.

The four configurations will predict the same untouched held-out records. McNemar's test will compare the proposed hybrid configuration with each baseline using paired correctness outcomes at $\alpha = .05$. The exact test statistic and p-values will be reported, and statistical significance will be interpreted together with Macro F1, false-positive rate, inference time, and practical feasibility. The study will not assume that the hybrid must outperform the baselines.

Credible-Source Retrieval

Serper.dev will provide the live search interface for related-source retrieval (Serper.dev, n.d.). The backend will apply the approved source and relevance policy to returned links; search ranking itself is not a credibility judgment. Query formation, source eligibility, and the handling of irrelevant, missing, or unavailable results will be documented and tested. The user will see contextual support separately from the prediction, confidence, and disclaimer.

Retrieval and classification will remain independent processes. A retrieval failure will not convert a prediction, and a strong retrieval match will not silently overwrite the model's class. The user may compare the result with the cited material and decide whether further verification is needed. This arrangement keeps the statistical evaluation of the classifier separate from the coverage of the search service.

Administrative evidence search follows the same principle with a different purpose. Reviewers may use the Google Fact Check Tools API to locate professional fact checks during dataset validation, but a returned rating is not copied automatically into the SuriPH

label. The reviewer must confirm claim correspondence and record the source, match decision, notes, and final validation outcome.

Controlled Dataset and Model Maintenance

The Admin Dashboard will distinguish candidate records from approved dataset items. Authorized users can examine provenance, language, claim group, evidence, annotations, and status before approving membership in a dataset version. Bulk import will not bypass validation. Audit records will identify important administrative actions and their outcomes.

Retraining creates a candidate model rather than replacing the active model immediately. The system will record the dataset version, configuration, metrics, artifact location, and approval state. An authorized evaluator or administrator can review the candidate and approve or reject activation. If rejected, the active version remains unchanged.

This controlled workflow protects the research from unreviewed feedback loops. Public submissions, browsing activity, prediction logs, or retrieved sources will not automatically become labeled training data. Any later use of public material would require the same approved screening and evidence process as other candidates, together with the required privacy and ethics controls.

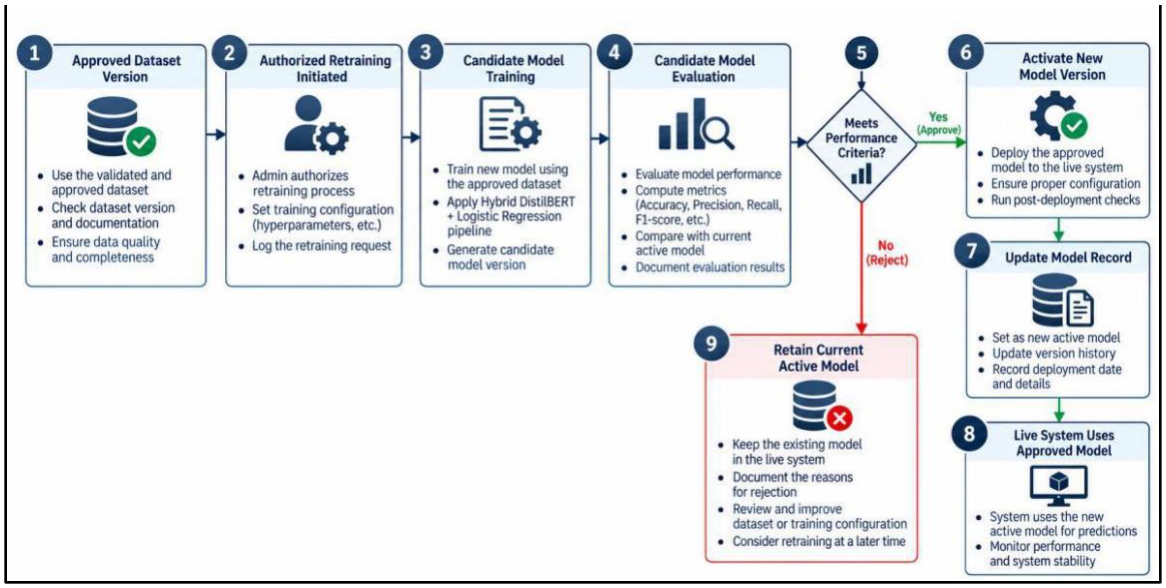


Figure 3.3 Controlled Model Retraining and Activation Flow

Methodology

The software component of the study will follow an Agile Scrum-based iterative development process using sprint planning, development, review, and retrospective cycles (Schwaber & Sutherland, 2020). Research controls—including evidence annotation, claim-group-aware partitioning, training-only fitting, held-out testing, and manual model approval—will operate alongside the sprints and will not be altered merely to fit a software iteration. This wording identifies the recognized development approach without presenting a researcher-created Hybrid Agile method as a separate established SDLC.

planning, review, and retrospective practices described by Schwaber and Sutherland (2020); the entire research methodology is not claimed to be Scrum.

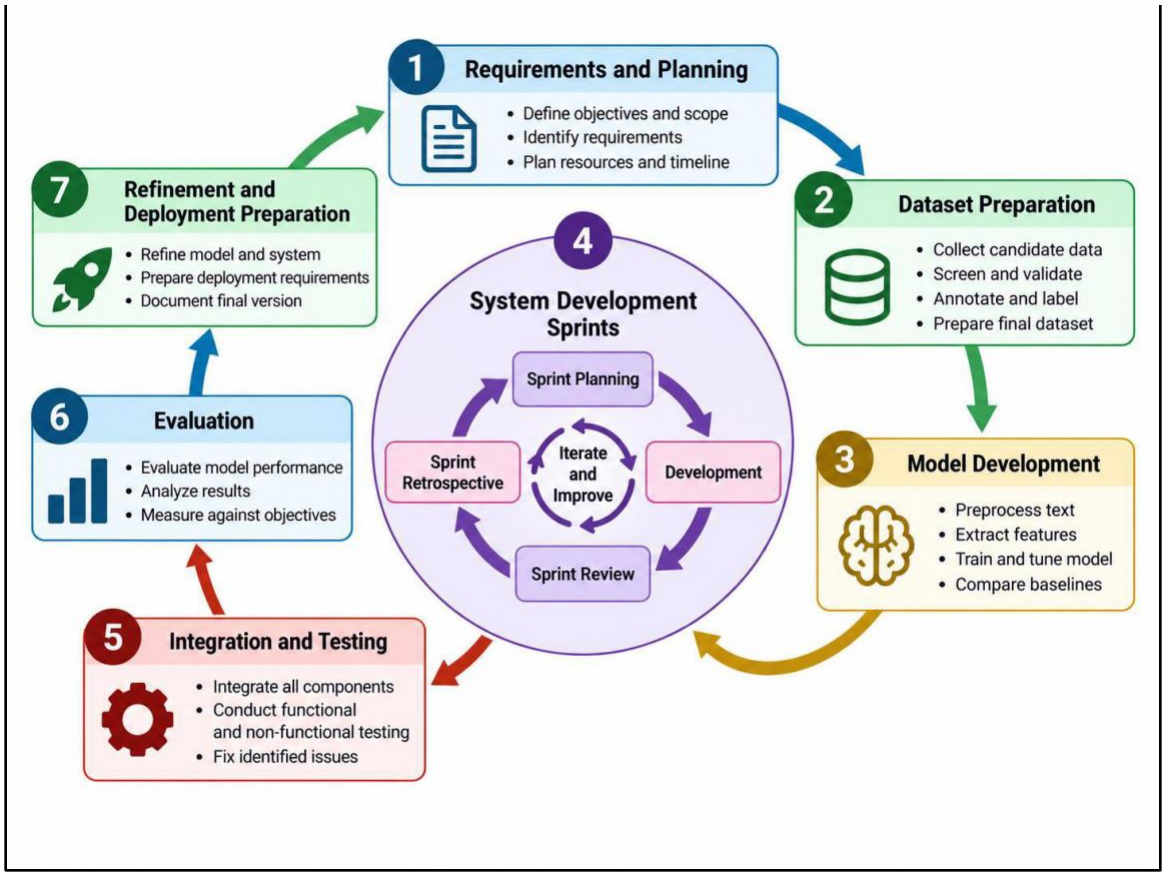


Figure 3.4 Agile Scrum-Based Iterative Development Process of SuriPH

Phase 1 Requirements and Planning

The researchers will confirm the finalized scope, users, functional and non-functional requirements, data-governance boundaries, evaluation design, and implementation-dependent decisions. AFP IT coordination and the separate general-user respondent plan will be documented before formal evaluation activities begin.

Phase 2 Dataset Preparation

Candidate records will be inventoried, Taglish-screened, grouped by claim, and reviewed against credible or authoritative evidence under the approved annotation guide. Only approved Taglish records will enter a traceable dataset version.

Phase 3 Model Development

The four approved configurations will use the same approximately 70% training, 15% validation, and 15% testing partitions with claim-group separation.

Training-data-only fitting, validation-based selection, and final held-out evaluation will preserve comparability and reduce leakage.

Phase 4 System Development Sprints

The browser extension, restricted Admin Dashboard, backend services, media-to-text pipeline, classifier, retrieval component, dataware, and model- version controls will be implemented iteratively. Each cycle will follow this exact sequence: Sprint Planning → Development → Sprint Review → Sprint Retrospective → Repeat.

Phase 5 Integration and Testing

The researchers will connect browser-side acquisition to backend processing and verify functional requirements across direct text, OCR, transcription, inference, retrieval, storage, authorization, controlled retraining, candidate-model review, and error handling.

Phase 6 Evaluation

Evaluation will comprise dataset composition and independent pilot agreement; held-out performance of the four configurations; representative functional and integration test outcomes, including optional media-to-text acquisition; AFP IT technical assessment; and a separate fourth-year BSIT general-user assessment. Macro F1 will remain the primary classifier measure. AFP IT and BSIT questionnaire results will never be pooled.

Phase 7 Refinement and Deployment Preparation

Documented defects, evaluator findings, and panel recommendations will guide final refinement. Deployment preparation will identify only the browsers, hardware, workload behavior, response times, and processing limits demonstrated by the completed prototype, and activation will remain restricted to an approved model version.

Requirement Analysis

Requirement analysis defines what SuriPH must do and the conditions under which it should operate. Functional requirements describe system behavior.

Non-functional requirements describe quality, security, privacy, compatibility, performance, and maintenance constraints.

Table 3.2 Functional Requirements of SuriPH

ID	Requirement Description	Priority
FR1	Allow a user to initiate a screening request for selected supported social-media post content encountered through a supported webpage.	High
FR2	Inspect supported regular DOM and accessible/open Shadow DOM through browser-extension content scripts within granted permissions.	High
FR3	Obtain directly accessible post text, captions, and selected contextual comments from supported webpage elements.	High
FR4	Extract visible text from supported images or screenshots through OCR.	High

ID	Requirement Description	Priority
FR5	Obtain spoken transcripts and sampled-frame visible text from supported webpage videos.	High
FR6	Allow optional submission of selected content to CandidateSamples for human review only, without assigning a final label or triggering retraining.	High
FR7	Combine relevant extracted textual content, including video transcript and on-screen OCR text where applicable, normalize/deduplicate it as required, and prepare it for feature extraction and classification.	High
FR8	Generate contextual representations using multilingual DistilBERT.	High
FR9	Calculate the finalized linguistic and stylistic features.	High
FR10	Fuse contextual and linguistic representations.	High
FR11	Classify the fused representation through Logistic Regression.	High
FR12	Produce an internal class prediction and confidence value.	High
FR13	Present Potentially Misleading or No Misleading Information Detected with the model confidence value and an assistive disclaimer.	High
FR14	Use Serper.dev to retrieve related information under the approved credible-source policy independently of classification, without recalculating or overwriting the prediction.	High
FR15	Display related context and credible-source links when available.	High

ID	Requirement Description	Priority
FR16	Communicate between the extension and backend through the implemented API.	High
FR17	Provide understandable errors for extraction, communication, inference, and retrieval failures.	Medium
FR18	Restrict the Admin Dashboard to authorized users.	High
FR19	Allow authorized candidate and evidence review, with Google Fact Check Tools API assisting dataset evidence retrieval only and human confirmation of claim correspondence.	High
FR20	Allow administrators to approve, revise, or exclude candidate records.	High
FR21	Maintain approved dataset versions and membership records.	High
FR22	Allow authorized administrators to initiate controlled retraining.	High
FR23	Display candidate-model evaluation results before activation.	High
FR24	Allow an authorized decision to approve or reject a candidate model.	High
FR25	Activate only an approved model version for live inference.	High
FR26	Maintain audit and version records for significant dataset review, retraining, candidate-model approval/rejection, and activation actions; a dedicated record-viewing screen is not assumed.	Medium

Table 3.3 Non-Functional Requirements of SuriPH

ID	Requirement Description	Priority
NFR1	Produce results within response times established by actual prototype benchmarking.	High
NFR2	Use HTTPS for deployed client-to-backend communication.	High
NFR3	Keep API keys, database credentials, and administrative secrets out of browser-side code.	High
NFR4	Minimize unnecessary collection and retention of browsing information.	High
NFR5	Remove temporary media after processing unless retained for approved research testing.	High
NFR6	Handle unavailable media, OCR failure, transcription failure, retrieval failure, and backend errors without uncontrolled failure.	High
NFR7	Use modular backend and machine-learning components to support testing and maintenance.	Medium
NFR8	Keep dataset and model versions traceable for reproducibility and controlled retraining.	High
NFR9	Present public results in language understandable to non-technical users.	High
NFR10	Restrict administrative functions to authorized personnel.	High
NFR11	Claim browser compatibility only for tested extension environments.	High

ID	Requirement Description	Priority
NFR12	Support expansion of validated records without redesigning the classification pipeline.	Medium
NFR13	Maintain stable operation during repeated requests within tested capacity.	High
NFR14	Keep retrieval failure from changing or invalidating the model prediction.	High
NFR15	Prevent raw public submissions from automatically becoming training data or triggering retraining.	High

Requirement analysis connects the objectives to observable system behavior. Each functional requirement identifies an action or result that can be demonstrated through an approved test case. Non-functional requirements identify conditions such as security, privacy, compatibility, reliability, maintainability, and performance that influence how the function operates. Priority labels support planning but do not replace acceptance criteria.

The public workflow begins with user activation and selection of supported social-media post content. It includes text acquisition, requested media-to-text processing, prediction and confidence, independent Serper.dev retrieval, cautious labels, and a disclaimer. Optional submission creates a candidate record only. Test cases will distinguish incomplete extraction, failed screening, and completed classification with unavailable retrieval.

The administrative workflow requires stronger controls. Only authorized users may review candidates, record evidence, manage validation decisions, create dataset

versions, initiate retraining, inspect candidate-model metrics, approve or reject a model, and activate an approved version. Important actions must remain traceable so that the team can determine which dataset and model produced a result.

The requirements will be updated only through the approved development process. If implementation testing shows that a media type, browser structure, or performance threshold is not feasible, the change will be documented in the scope and test plan. The final manuscript will describe the system actually evaluated rather than an earlier design that was not completed.

Population, Sample Size and Sampling Techniques

Population

The study will use two distinct evaluator populations. The general-user evaluation population will consist of eligible fourth-year Bachelor of Science in Information Technology students from the selected class block. Each eligible student must be at least 18 years old and able to interact with English, Filipino, or Taglish online content. The separate technical-evaluator population will consist of qualified and authorized AFP Information Technology Department personnel whose work or training is relevant to software, networking, security, databases, system administration, maintenance, or related information-technology functions.

Sample Size

All eligible members of one selected fourth-year BSIT class block will be invited through total enumeration within that accessible block. The final report will state the eligible, invited, and completed-response counts. The class block is an accessible

general-user group, not a statistically representative sample of all Filipino internet users; conclusions from its responses will therefore be limited to prototype interaction and acceptability within that group. AFP IT technical evaluators will be selected separately through criterion-based purposive sampling according to relevant IT role, experience, authorization, and availability. No Slovin formula will be used.

The formula is not applied to AFP IT personnel and does not make a selected class block representative of the general public.

Sampling Techniques

If all eligible students in the selected class block are invited as the intended respondent group, the procedure will be described as total enumeration or a census within that block, and Slovin's formula will not be presented as necessary.

If only a sample is selected, the final sampling procedure will match the adviser- and statistician-approved design. AFP IT technical evaluators will be selected separately through the approved purposive procedure because participation requires relevant expertise and institutional authorization. Their number will not be combined with the general-user population or sample-size computation.

Results from AFP IT personnel and fourth-year BSIT general-user respondents will be analyzed and reported separately because the groups perform different tasks and evaluate different aspects of the prototype.

Description of Respondents

AFP IT respondents will evaluate the completed prototype through tasks that expose the implemented technical behavior. Depending on authorization and final scope,

they may examine extension-to-backend communication, error handling, authentication, administrative access, model-version information, performance, compatibility, or maintenance documentation. Their questionnaire will include only quality characteristics that can be assessed from the system and provided materials.

Eligible fourth-year BSIT respondents will use the extension through a guided set of representative tasks. They will evaluate whether activation is understandable, whether system status is visible, whether results and confidence are explained clearly, whether credible links are accessible, and whether the disclaimer communicates the limits of the prediction. Technical jargon will be avoided in items intended for this group.

Both groups will receive consistent instructions and an opportunity to report problems or observations. Demographic or profile information will be limited to variables required by the approved research design. Participation records and responses will be protected according to institutional and privacy requirements.

Research Instrument

Dataset Annotation and Evidence Validation Form The Dataset Annotation and Evidence Validation Form will record sample identifier, text, language, topic, provenance, original or preliminary label, claim_group_id, evidence source and publisher, URL, rating where applicable, claim-match decision, reviewer identifier, independent judgment, validation date, notes, and Approve, Revise, or Exclude status. Independent judgments and the final resolved Misleading or Non-Misleading label will occupy separate fields.

Insufficient evidence is not equivalent to evidence that contradicts a claim; unresolved cases may be revised or excluded.

The annotation guide will accompany the form with inclusion, exclusion, segmentation, and disagreement rules. Reviewers will receive examples of checkable and non-checkable content, sufficient and insufficient context, duplicate claims, and acceptable evidence. A pilot annotation round will identify ambiguous instructions before the full validation process begins.

Model Evaluation Record Each model experiment will have a record linked to its dataset version and fixed partition identifiers. Once implementation is frozen, the record will specify the actual checkpoint and tokenizer revision, frozen or fine-tuned mode, pooling, long-text handling, contextual dimensions, feature definitions and scaling, Logistic Regression settings, threshold, software versions, seeds, and artifact location. It will then capture observed confusion matrices, class-level metrics, Macro F1, False Positive Rate, subgroup results, bootstrap settings and intervals, and paired comparisons. Unimplemented choices will not be filled with assumed values.

Candidate-model review will use the experiment record and documented acceptance procedure. Reviewers will distinguish validation evidence used for selection from held-out evidence used for final reporting. Repeated maintenance cycles must not tune a candidate against the same final test outcomes. Approval and activation will be recorded separately, and an increase in one aggregate score alone will not establish acceptable false-positive behavior.

Media-to-Text Functional Test Record The record will document the selected media case, expected verified text, observed output, system status, and failure notes. These representative checks establish whether the optional acquisition feature operates

acceptably under the tested conditions; they do not form a separate OCR or speech-recognition accuracy experiment.

Extraction cases will represent the media conditions supported by the prototype. The researchers will avoid using only clean laboratory examples if the public system is intended to process screenshots or webpage media. At the same time, unsupported or deliberately adversarial cases will be identified as such so that the reported evaluation matches the implemented scope.

Functional and Integration Test Cases Each functional and integration test case will include its Requirement ID and corresponding Objective reference, prerequisites, input or action, expected result, actual result, status, and version tested. Failed cases will include reproducible notes and a linked retest instead of replacing the failure record. The suite will cover FR1–FR26 and applicable NFR1–NFR15, including candidate- only submission, failure handling, authorization, and retrieval independence.

Integration tests will verify data flow across component boundaries.

Examples include sending extracted webpage text to the backend, connecting OCR or transcription output to preprocessing, returning active-model information with a prediction, keeping retrieval results separate, applying authentication to administrative endpoints, and preserving dataset-to-model traceability during retraining.

Software Quality Evaluation Questionnaire

Table 3.4 Proposed Agreement Scale

Scale	Response
5	Strongly Agree

Scale	Response
4	Agree
3	Neutral
2	Disagree
1	Strongly Disagree

Separate technical and general-user questionnaires will use a proposed five-point agreement scale. Items will be developed from the applicable, observable characteristics of ISO/IEC 25010:2023 and reviewed for relevance and clarity before use (International Organization for Standardization, 2023). The standard supplies a quality model, not a ready-made questionnaire.

Characteristics that cannot be assessed through the approved procedure will not be included merely for completeness.

A five-point agreement scale provides a consistent response format, but numerical interpretation ranges will be approved with the statistician. Results will be summarized by item and applicable characteristic. Open comments may supplement the scale by explaining a defect or usability concern that a mean score cannot describe.

Technical and general-user questionnaires will share only items that genuinely apply to both groups. Security configuration, maintainability, or administrative traceability may be appropriate for technical personnel, while ease of requesting screening and understanding the result belong to general-user assessment. Separate instruments reduce the pressure to ask respondents about matters outside their experience.

Data Gathering Procedures

Step 1

Approval and coordination Before collection or evaluation, the researchers will seek the required adviser, institutional, ethics-related, and statistical approvals for the instruments, respondent plan, data handling, and testing procedures. Coordination with AFP IT will confirm authorized participants, environment, and demonstration access.

Academic evaluation will not imply permission to connect the prototype to protected institutional resources.

Step 2

Candidate data preparation Candidate sources will be inventoried with their release, purpose, permitted use, original label meaning, language, topic, and provenance. The team will screen contextual completeness and checkability and document repairs, anonymization, duplicates, and exclusions. Formal-news and social-media records will remain distinguishable. This step produces candidate records, not final ground truth.

Step 3

Evidence retrieval and human validation Following a pilot annotation round, reviewers will apply the approved guide and preserve independent judgments on the designated overlap subset. Google Fact Check Tools API may locate published reviews, but reviewers must confirm claim correspondence and document evidence, publisher, rating where applicable, URL, decision, and notes. Approved credible or authoritative sources may be used when no suitable fact check exists. Disagreements will be resolved only after the independent records needed for Fleiss' Kappa have been retained; uncertain records may be revised or excluded.

Step 4

Partitioning and model development Validated records will receive claim_group_id before approximately 70% training, 15% validation, and 15% testing assignment. Equivalent claims will remain together, with stratification where compatible with grouping. All four configurations will use the same partitions. Data- dependent preprocessing and transformations will be fitted only on training data, while validation data will support configuration and threshold selection before final testing.

Step 5

Held-out model evaluation. After configuration selection is frozen, the four models will predict the same untouched held-out records. The researchers will compute Macro F1, class-level precision, recall and F1-score, accuracy, confusion matrices, false-positive rate, and inference time. McNemar's test will compare the paired correctness outcomes of the proposed hybrid and each baseline. Test-set error analysis will describe limitations and will not be used to tune or reselect the reported model.

Step 6

Media-to-text functional testing. Representative supported images, screenshots, and bounded video segments will be tested against manually verified expected text. The record will document success, incomplete extraction, empty output, unsupported or protected media, timeouts, and clear user-facing error status. These checks establish functional feasibility and observed limitations; CER and WER are not separate study metrics and acquisition outcomes will not be combined with classifier-performance measures.

Step 7

Functional and integration testing The researchers will execute the approved functional and integration suite, linking each case to its Requirement ID and Objective reference. Evidence will record outcomes, failure details, and corrected retests for both public and administrative workflows. Tests will include retrieval failure after completed classification, candidate-only submission, administrative access, controlled retraining, manual approval, and approved-model activation without exposing real credentials or protected data.

Step 8

Technical and general-user evaluation Qualified and authorized AFP IT personnel and eligible fourth-year BSIT general-user respondents will receive consent information, orientation, approved tasks, and the questionnaire appropriate to their role. A consistent prototype version and documented environment will be used where feasible. Eligible, invited, and participating counts will be recorded without assuming full participation. Responses will be checked for completeness without pressure to change judgments, and the two groups will be analyzed separately.

Step 9

Compilation and protection of data Research records will be stored under controlled identifiers with access restricted to authorized research roles. Collection will be limited to data needed for the approved purpose; unrelated browsing and personal information will not be retained. Temporary media and derivatives will be deleted after processing unless specifically retained for approved research testing, in which case an approved retention and deletion schedule will apply. Respondent information will be

separated from analysis summaries. Public submissions, inference records, and retrieved links will never become approved labels or training data automatically.

The final analysis will keep raw observations separate from interpreted summaries. Corrections to data or coding will be documented. Tables and manuscript statements will use the same approved counts and version identifiers so that a reader can trace reported results to the underlying evaluation records.

Statistical Treatment of Data

The study will apply descriptive, agreement, predictive-performance, extraction-quality, uncertainty, and paired-comparison procedures according to the finalized research design. Technical-evaluator and general-user questionnaire results will remain separate. Model measures will be calculated from held-out predictions, while OCR and transcription error measures will use verified reference text when available. Macro F1 will serve as the primary model-performance measure.

Percentage. Percentage will summarize frequencies such as class, language, topic, provenance, exclusion reason, respondent profile, and test-case outcome. Every percentage for a subgroup will be reported with its denominator.

$$P = \frac{f}{N} \times 100$$

P is the percentage, f is the frequency of the category, and N is the total number of observations in the relevant group.

Fleiss' Kappa will measure chance-corrected agreement among the four researchers on the overlapping pilot subset before any reconciliation (Fleiss, 1971). Only the independent binary proposed labels will enter the calculation. Evidence sufficiency

and management decisions such as Revise or Exclude will be documented separately. The actual κ value will be reported; κ = .61-.80 will be described as substantial agreement under the cited interpretation guideline, while agreement will not be presented as factual validity.

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

Here, κ is agreement beyond chance, $\bar{P}$ is the mean observed agreement, and $\bar{P}e$ is the chance agreement from the overall category proportions. For M independently rated records, r ratings per record, and k approved nominal categories, n_{ij} denotes the number of ratings in category j for record i. The detailed calculation will follow the approved reviewer design and analysis procedure.

Accuracy. Accuracy will describe the overall proportion of held-out records classified correctly (Rainio, Teuho, & Klén, 2024).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

TP and TN are correctly predicted Misleading and Non-Misleading records, while FP and FN are the corresponding errors. All counts refer to the same held-out subset.

Precision. Precision will indicate the proportion of records predicted as Misleading that are labeled Misleading in the held-out data.

$$Precision = \frac{TP}{TP + FP}$$

TP is the number of true-positive Misleading predictions, and FP is the number of Non-Misleading records incorrectly predicted as Misleading.

Recall. Recall will indicate the proportion of held-out Misleading records detected by the classifier.

$$Recall = \frac{TP}{TP + FN}$$

TP is the number of true-positive Misleading predictions, and FN is the number of Misleading records incorrectly predicted as Non-Misleading.

F1-Score. The class-level F1-Score will combine precision and recall through their harmonic mean (Rainio et al., 2024).

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Precision is the positive predictive value for the class, and Recall is the sensitivity for that class.

Macro F1. Macro F1 will average the class-level F1 values without allowing the larger class to dominate and will serve as the primary measure for comparing the four configurations.

$$Macro\ F1 = \frac{F1_{Misleading} + F1_{Non-Misleading}}{2}$$

F1Misleading is the F1-Score for the Misleading class, and F1Non-Misleading is the F1-Score for the Non-Misleading class.

False Positive Rate. False Positive Rate will be examined separately because an incorrect warning about legitimate material may reduce trust and cause unnecessary concern.

$$FPR = \frac{FP}{FP + TN}$$

FPR is the False Positive Rate, FP is the number of Non-Misleading records predicted as Misleading, and TN is the number of Non-Misleading records classified correctly.

Weighted mean will summarize valid responses to the approved five-point questionnaire items, with weights 5, 4, 3, 2, and 1 assigned from the most favorable to the least favorable response. Item and characteristic means will be interpreted using the stated equal-width ranges in Table 3.3. Because AFP IT personnel perform technical evaluation and BSIT students perform general-user evaluation, their results will be computed and reported separately; no combined grand mean will be produced. The final questionnaire will be reviewed by the adviser and statistician before administration.

$$\bar{x} = \frac{\sum fx}{N}$$

x̄ is the weighted mean, Σfx is the sum of each response frequency multiplied by its assigned weight, and N is the total number of valid responses in the relevant evaluator group.

McNemar's Test. McNemar's test will compare the paired held-out predictions of the proposed hybrid model with each comparison configuration. The planned significance level is α = 0.05 unless an approved analysis plan specifies otherwise (McNemar, 1947).

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

Here, b counts cases correctly classified by the first model and incorrectly by the second, while c counts the reverse. The displayed statistic is the continuity-corrected form and is used only when its assumptions are appropriate.

System Requirements

Hardware

Table 3.5 Hardware Requirements

Component	Requirement
Development and training workstation	CPU, RAM, storage, and optional GPU acceleration sufficient for transformer experiments, OCR, Whisper, local integration, and dataset preparation. Final specifications will match the equipment actually used.
Backend server	Resources sufficient for FastAPI, active-model inference, OCR, transcription, retrieval coordination, temporary processing, database access, and the tested request capacity.
End-user device	A computer capable of running a supported browser and extension. Heavy model processing is expected to occur on the backend.
Administrator device	A secured computer with a supported browser and authorized access to the Admin Dashboard.

Hardware requirements will be based on the completed implementation and measured workloads. The development and training workstation must support dataset preparation, repeated transformer representation generation, OCR, transcription, and model evaluation. GPU acceleration may reduce experiment time, but the final specification will distinguish a convenience for development from a true minimum requirement.

The backend server must support active-model inference, request coordination, temporary media processing, retrieval, structured database access, and the tested level of concurrent use. Storage planning must account for model artifacts, approved dataset exports, logs permitted by the research design, and temporary files. Capacity claims will be tied to the workload used during benchmarking.

End-user and administrator devices require a supported browser and network access to the backend. Heavy model work is expected to occur on the server rather than on the user's computer. The final browser and operating- system combinations will be named only after actual compatibility testing.

Software and Technology Stack

The SuriPH technology stack separates browser-side presentation and webpage inspection from backend orchestration, machine learning, media processing, retrieval, database operations, and model storage. The React-based browser extension will use content scripts and WebExtensions-compatible APIs to inspect regular DOM and accessible or open Shadow DOM structures within granted permissions. The backend will not inspect the webpage DOM directly. It will receive the resulting text or media-processing request and coordinate heavy or security-sensitive processing. dependency

has already been deployed. Optional icon and server-runner choices are omitted because the supplied manuscript does not confirm their use. Before implementation testing, this inventory will be reconciled with the actual dependency files and build records; unused tools will be removed and retained versions documented.

Table 3.6 Planned Software and Technology Stack

System Layer	Technology	Role in SuriPH
Frontend Framework	React	Main user-interface framework for the browser extension and restricted Admin Dashboard.
Frontend Build Tool	Vite	Development server and production build tooling for the React frontend.
Frontend Styling	Tailwind CSS	Layout, responsive styling, and consistent interface presentation.
Browser Extension Integration	WebExtensions-compatible APIs	Content scripts, permissions, messaging, regular DOM inspection, and accessible/open Shadow DOM inspection.
Backend Language	Python	Primary server-side and machine-learning programming language.
Backend Framework	FastAPI	REST/API endpoints and orchestration of backend processing.
Transformer Framework	Hugging Face Transformers	Loads the tokenizer and multilingual DistilBERT model.

System Layer	Technology	Role in SuriPH
Contextual Model	distilbert-base-multilingual-cased	Produces contextual text representations for Taglish, Filipino, and English inputs.
Deep-Learning Runtime	PyTorch	Executes transformer-model operations.
Traditional Machine Learning	scikit-learn	TF-IDF baseline, Logistic Regression, transformations, and evaluation utilities.
Data and Feature Processing	pandas and NumPy	Dataset preparation, numerical operations, and implemented linguistic/stylistic feature processing.
Optical Character Recognition	EasyOCR	Extracts visible text from supported images, screenshots, and sampled video frames.
Speech-to-Text	Whisper	Transcribes spoken content from supported webpage videos.
Media Processing	FFmpeg	Audio extraction, media conversion, and video-frame sampling.
Structured Database	MySQL / MariaDB	Stores administrative, dataset, evidence, annotation, dataset-version, model-version, and audit metadata.
File and Model Storage	Server filesystem	Stores trained model artifacts, tokenizer resources, approved dataset exports, and temporary media.

System Layer	Technology	Role in SuriPH
Dataset Evidence Retrieval	Google Fact Check Tools API	Assists human reviewers in locating published fact-check records during dataset validation.
Live Credible-Source Retrieval	Serper.dev	Retrieves related credible-source results independently from the classifier.
Version Control	Git / GitHub	Tracks source-code revisions and project versions.

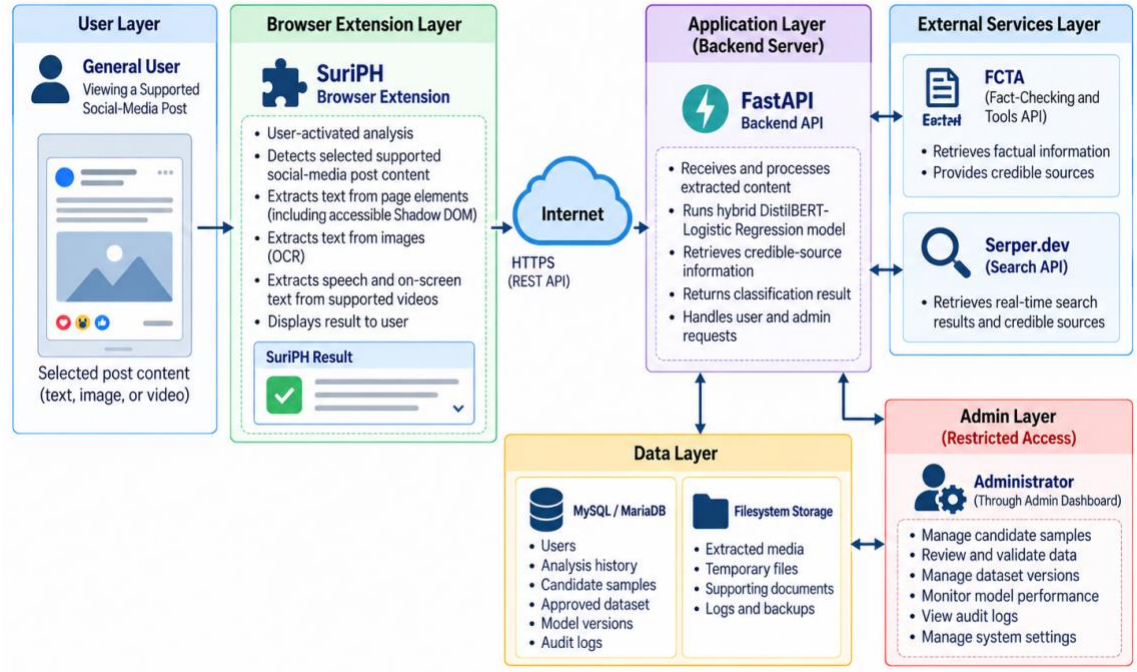


Figure 3.5 System Architecture of SuriPH

In Figure 3.5, the extension is responsible for selecting and obtaining supported social-media post content; the backend receives text or approved media-processing requests. FCTA denotes Google Fact Check Tools API and is used only for dataset evidence retrieval under human review. Serper.dev supplies independent live source retrieval. Generic Users in the storage layer means restricted AdminUsers, while any analysis history is optional InferenceLogs under the approved privacy design, not a public-account or compulsory browsing-history service. Model and media artifacts remain on the protected filesystem. The diagram's broad labels do not authorize functions beyond the requirements.

Peopleware

Table 3.7 Peopleware Requirements

Role	Responsibility
Researchers and developers	Prepare data, implement and test the model and system, maintain documentation, and conduct the approved evaluation.
Research adviser	Guides the research design, scope, manuscript, and approval process.
Statistician	Reviews sampling, instruments, metrics, statistical tests, and interpretation procedures.
Dataset reviewers or annotators	Review claims and evidence, record decisions, and participate in disagreement resolution.

Role	Responsibility
AFP IT technical evaluators	Assess applicable technical and software quality characteristics of the prototype.
Eligible fourth-year BSIT general-user evaluators	Assess public-facing interaction, result presentation, credible links, disclaimer clarity, and error feedback.
Authorized administrator	Uses restricted dataset and model-management functions within the approved scope.

Peopleware includes the researchers and developers who prepare data, implement the system, run experiments, test functions, and maintain the manuscript. The research adviser guides scope and academic decisions, while the statistician reviews sampling, instruments, measures, statistical procedures, and interpretation. These roles help keep the research method aligned with approved academic and statistical requirements.

Dataset reviewers or annotators examine claims and evidence, record independent decisions, and participate in the approved disagreement process.

AFP IT technical evaluators assess applicable quality characteristics and technical operation. Eligible fourth-year BSIT general-user evaluators assess the public interaction. Authorized administrators manage only the restricted dataset and model functions allowed by the approved prototype.

Role separation will be enforced where practical. Public users will not gain administrative privileges, and evaluator participation will not automatically authorize model activation or dataset approval. The final documentation will identify which actions

require authentication, which require an approval decision, and which are automated backend processes.

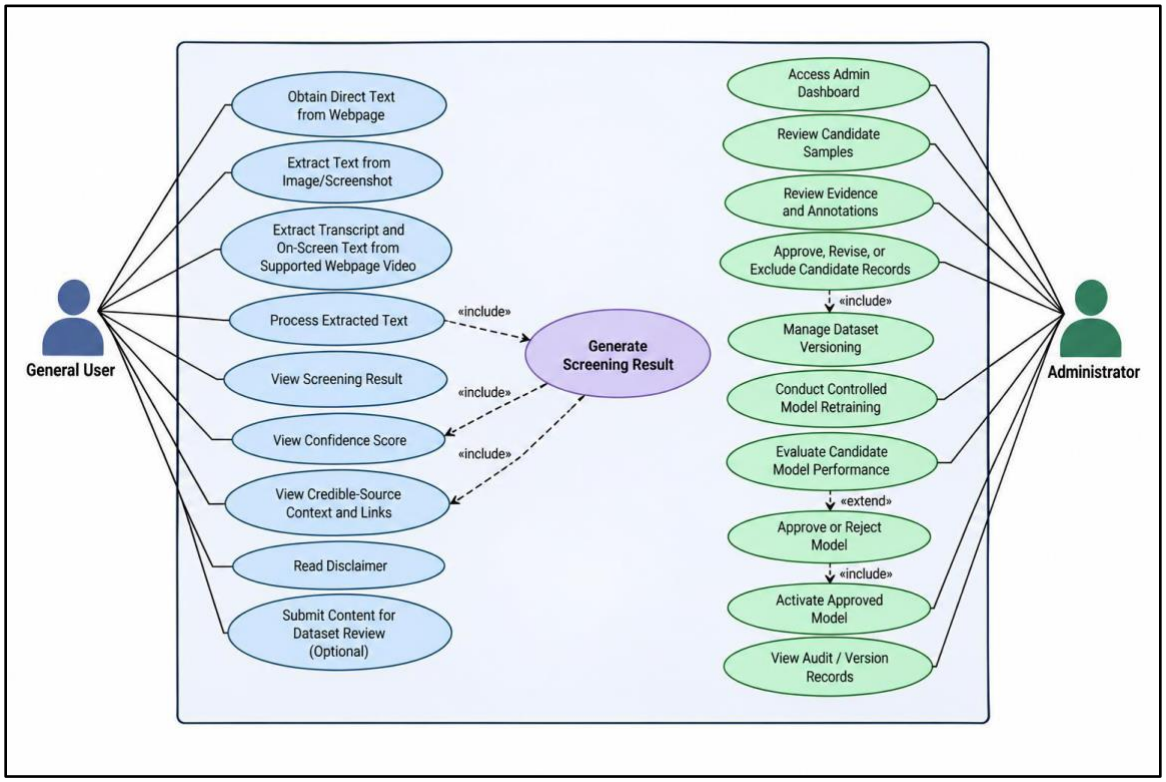


Figure 3.6 Use Case Diagram of SuriPH

supported social-media post content. The extension can obtain direct text or request supported image OCR, video transcription, and frame OCR. Relevant text is prepared for the same classifier. Results comprise the exact public label, model confidence, separate Serper.dev context and links when available, and a disclaimer. Optional submission creates CandidateSamples for human review; it does not assign a final label or trigger training.

The Administrator uses the restricted Admin Dashboard for candidate and evidence review, Approve, Revise, or Exclude decisions, approved dataset versions, authorized retraining, candidate-model evaluation, manual approval or rejection, and activation of an approved version. Google Fact Check Tools API assists evidence

retrieval only during validation. Audit and version records support traceability; the diagram's View Audit / Version Records label is not a claim of a dedicated viewing screen unless that screen is implemented and tested.

Network

The deployed academic prototype will require a network connection between the browser extension or restricted Admin Dashboard and the backend.

Application traffic will use HTTPS in the tested deployment environment. Browser-side code will not contain database credentials, administrative secrets, model artifacts, or third-party API keys.

The backend will coordinate inference, OCR, transcription, retrieval, database access, temporary file handling, and model-version selection. Server- side integrations will contact the structured database, server filesystem, Google Fact Check Tools API during human-supported dataset validation, and Serper.dev for planned live credible-source retrieval. Retrieval availability will not determine or modify the classifier output.

Administrative endpoints will require authentication and authorization.

Database and model-management services will not be exposed directly to the public extension. If retrieval fails after classification, the prediction may still be presented with a clear notice that related sources are unavailable.

Exact network performance, timeouts, rate limits, browser compatibility, workload capacity, and response behavior will be reported only for the environments and conditions actually tested.

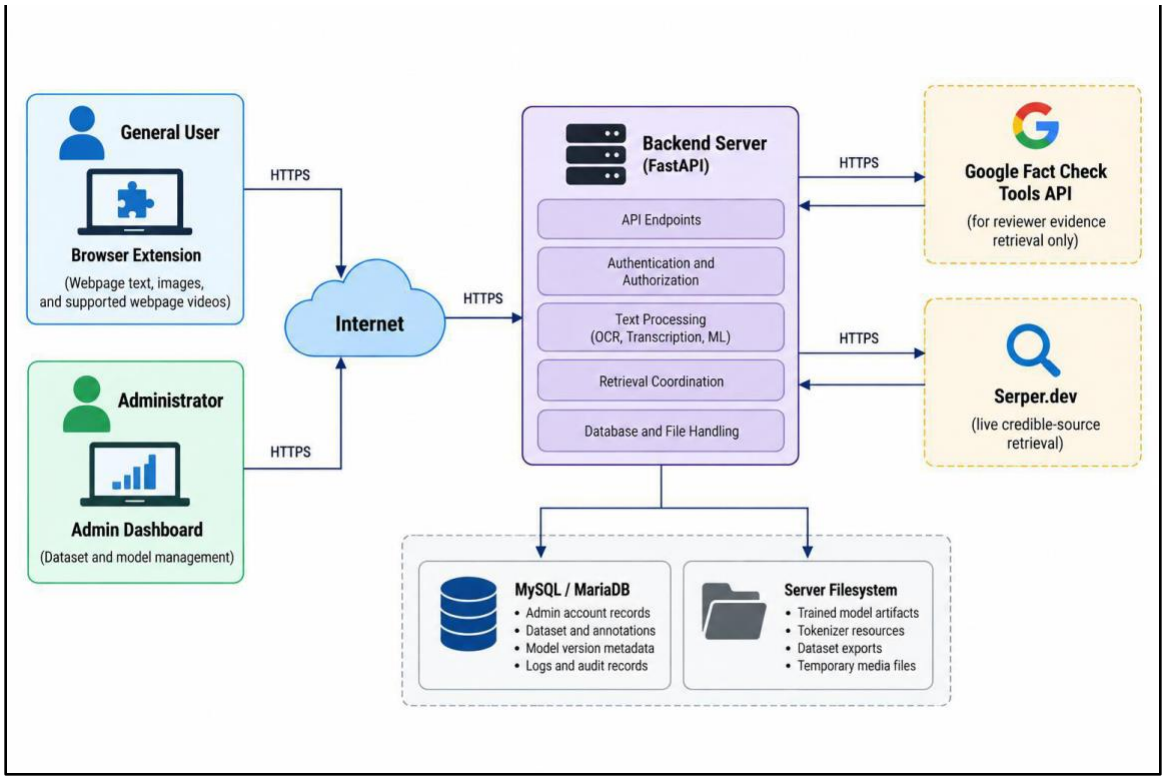


Figure 3.7 Network Architecture of SuriPH

Dataware

SuriPH will separate structured administrative and research metadata from file-based artifacts. MySQL or MariaDB will store structured records, while the protected server filesystem will store trained model artifacts, tokenizer resources, approved dataset exports, and temporary media required for approved processing or testing.

The main logical entities are AdminUsers, CandidateSamples, EvidenceRecords, Annotations, DatasetVersions, DatasetItems, ModelVersions, and AuditLogs. AdminUsers represent only authorized administrative accounts; the design does not assume public user accounts. CandidateSamples preserve text, provenance, language, topic, claim_group_id, a preliminary or proposed label, and validation status. EvidenceRecords preserve claim-match status, source or URL, rating where applicable,

reviewer or verification metadata, and notes. Annotations use the reviewer identifier established by the approved reviewer design and do not require every annotator to be an AdminUser.

DatasetVersions and DatasetItems will identify approved record membership, final labels, and partition assignments for each research cycle. ModelVersions will link each candidate or active model to the dataset version, actual configuration, measured metrics, approval and activation states, and filesystem artifact location.

AuditLogs will record significant authorized actions. Figure 3.8 is a logical design: physical field names, key constraints, credential protection, and the method for resolving artifact paths will be documented from the implemented schema rather than inferred as finished functionality.

InferenceLogs and RetrievedSources will be optional and will be enabled only if the completed implementation and approved privacy design require them. Raw public-user activity will not automatically become approved training data, and temporary media will not be retained beyond the approved processing or testing period. Physical field names, retention periods, backup procedures, and storage limits remain implementation-dependent until the completed prototype is verified.

Table 3.8 Planned Dataware Entities

Entity	Purpose
AdminUsers	Authorized administrative identities and access-control information.
CandidateSamples	Candidate text, language, topic, provenance, claim group, proposed label, and review status.

Entity	Purpose
EvidenceRecords	Evidence sources, links, professional fact-check information where applicable, and human claim-match decisions.
Annotations	Independent reviewer decisions, notes, dates, and approved disagreement-resolution information.
DatasetVersions	Traceable approved dataset releases and their composition metadata.
DatasetItems	Membership and claim-group-aware partition assignment of approved records within a dataset version.
ModelVersions	Model configuration, dataset version, evaluation results, artifact location, approval state, and activation state.
AuditLogs	Traceable records of significant authorized administrative actions and outcomes.

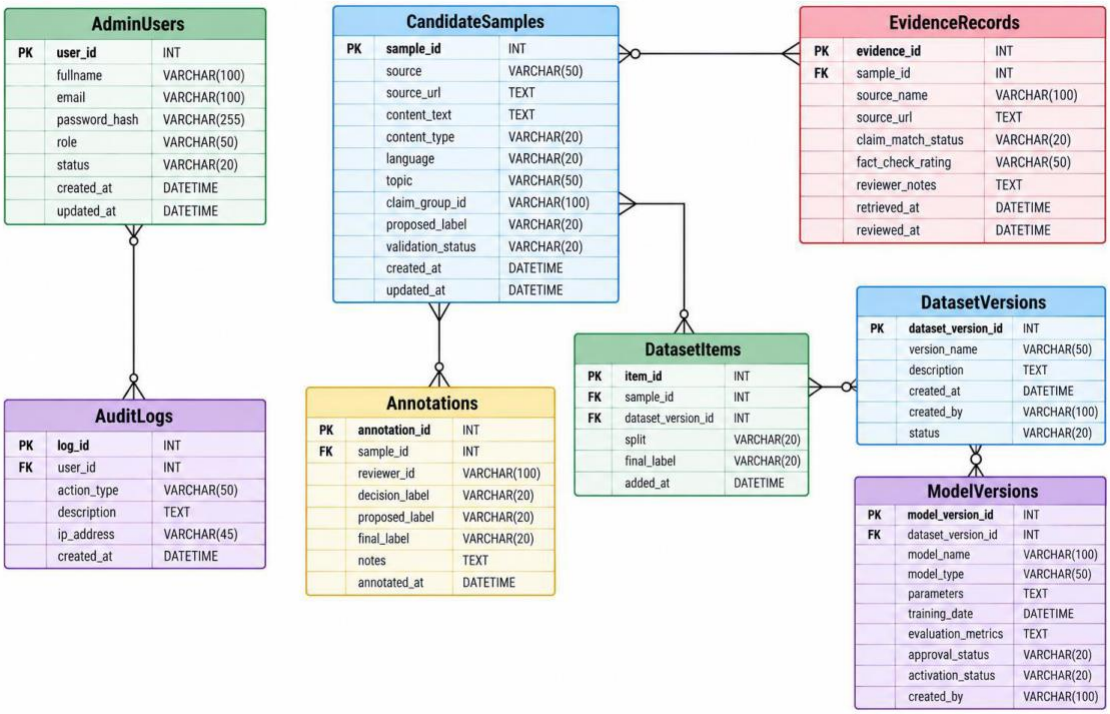


Figure 3.8 Entity Relationship Diagram of SuriPH

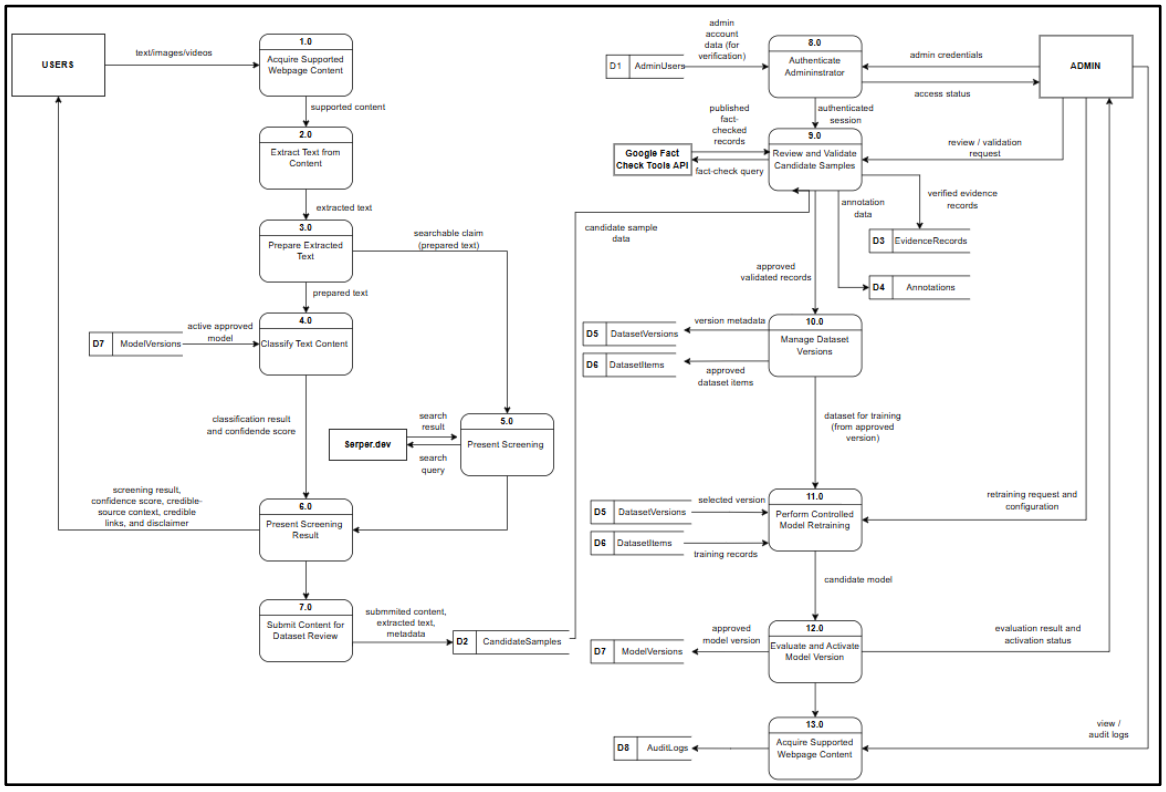


Figure 3.9 Data Flow Diagram Level 1 of SuriPH

User initiates screening of selected supported social-media post content. The extension obtains accessible post text or requests EasyOCR extraction from supported images and video frames and Whisper transcription from accessible video speech. Relevant transcript and OCR text are normalized and deduplicated before the active text classifier returns its prediction and confidence. The live retrieval branch uses Serper.dev to obtain related credible-source context; it does not modify, overwrite, or recalculate the prediction. A retrieval failure may leave the completed screening result available with an appropriate notice.

Optional submission is a separate action that creates candidate data for review, not approved training data. Authorized administrators review candidates and evidence, with Google Fact Check Tools API used only to locate published fact checks for human claim matching and validation. Approved records may enter a controlled dataset version. Retraining is manually initiated; candidate evaluation, manual approval or rejection, and activation of an approved model are distinct decisions even where the diagram compresses them into one process box. The lower audit-related branch represents administrative and version traceability, not a second public-content acquisition service. Public requests and submissions cannot automatically alter dataset labels or the active model.

‘

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